

Session 4.1

Aiyagari, Solved Twice: A Fixed Point, Then a Law of Motion

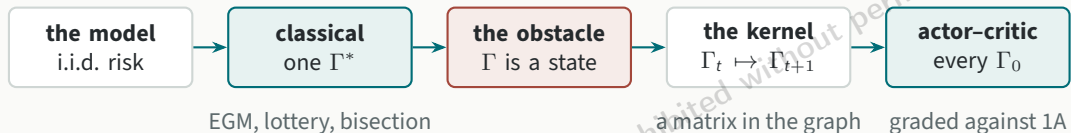
The classical solve returns one stationary distribution. An actor-critic solve with the distribution as its state returns how every distribution evolves



Zhigang Feng

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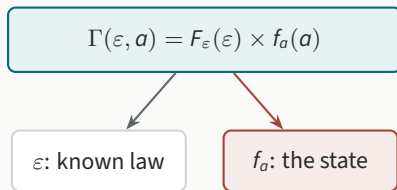
From a Fixed Point to a Dynamical System



- ▶ One economy throughout: Aiyagari with productivity redrawn each period (i.i.d.), no government, no aggregate risk.
- ▶ **Stage 1A**, classical: the stationary distribution Γ^* and the prices that support it.
- ▶ **Stage 1B**, networks that read the whole distribution: the map $\Gamma_{t+1} = \Phi(\Gamma_t)$, so the path from any start.
- ▶ **Stage 1A is the benchmark Stage 1B is graded against: same calibration, same quadrature.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Aiyagari, *QJE* (1994).

The Economy



ε is redrawn every period, independently of a , so today's assets tell nothing about today's ε

- ▶ $\log \varepsilon \sim N(0, 0.2^2)$, i.i.d.;
 $c + a' = (1 + r)a + w\varepsilon l, a' \geq 0$.
- ▶ Utility $(c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l})^{1-\sigma} / (1 - \sigma)$: no wealth effect on hours, so $l(\varepsilon) = (w\varepsilon / \psi_l)^{1/\theta_l}$.
- ▶ $Y = K^\alpha L^{1-\alpha}$; r (net of δ) and w from marginal products; $K = \int a d\Gamma$.
- ▶ $\Gamma_{t+1} = \Phi(\Gamma_t)$, and with i.i.d. ε the asset marginal f_a is the whole state.
- ▶ **Prices depend on Γ , so the saving rule is $g(\varepsilon, a; \Gamma)$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\beta = 0.95, \sigma = 2, \alpha = 0.34, \delta = 0.06, \theta_l = 0.32, \psi_l = 1.5$.

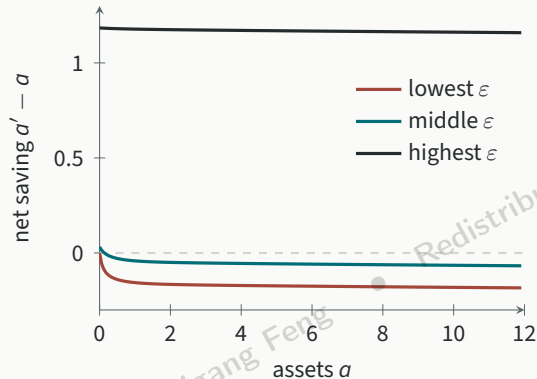
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Stage 1A: The Classical Solve

A household, a population and a price, in the benchmark every later number is graded against



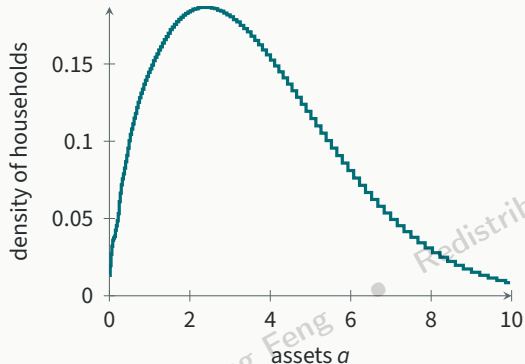
The Household: the Endogenous Grid Method



- ▶ Grid on tomorrow's a' ; invert the Euler equation for c ; today's a from the budget.
- ▶ Below the first recovered a the limit binds: $a' = 0$. A comparison, not a solver.
- ▶ 1500 asset nodes; 5 Gauss–Hermite nodes for ε , as the network uses later.
- ▶ **The lucky save, the unlucky dissave. The economics is the gap between these lines and zero.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Carroll, *Economics Letters* (2006).

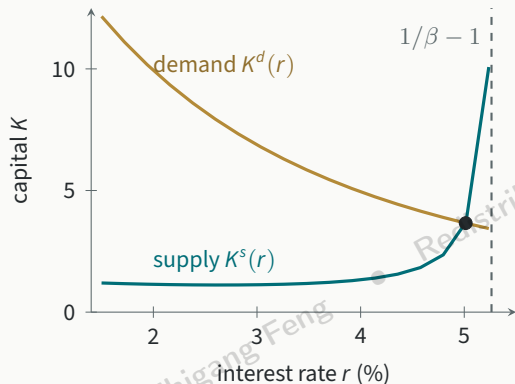
Young's Lottery



- ▶ Mass whose a' lands between nodes a_j, a_{j+1} is split: $\omega = \frac{a' - a_j}{a_{j+1} - a_j}$ up, $1 - \omega$ down.
- ▶ Mass is conserved and the mean of a' is exact: next period's K is exact, given the policy.
- ▶ Stationary λ : one dense solve of $(P_g^\top - I)\lambda = 0, \sum \lambda = 1$.
- ▶ Result: 0.009% at $a = 0$, Gini 0.338.
- ▶ **A histogram plus this lottery is the law of motion, as a matrix.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Young, *JEDC* (2010).

The Price: Bisection Inside a Proven Bracket



- ▶ Each supply point is one household solve and one distribution solve.
- ▶ Supply explodes as $r \rightarrow 1/\beta - 1$: the bracket's top. Prove a sign change before bisectioning.
- ▶ $r^* = 5.014\%$, $K^* = 3.6595$, $K/Y = 3.09$; about 30 CPU-seconds.
- ▶ $\beta(1 + r^*) = 0.99763$. **That number, so close to one, returns in every section that follows.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Aiyagari (1994).

Is the Benchmark Right?

Check	Measured	What it rules out
a second algorithm: value iteration	gap 0.0489 at 400 nodes, 0.0251 at 800: halves	an error in the EGM code
grid refinement, 800 against 1500 nodes	K moves by 0.003%	a grid that sets the answer
a panel of 200 000 households	mean drifts 1.4×10^{-5} in 1000 periods	an error in the lottery
Euler equation, with the limit market clearing at r^*	● worst interior error 5.9×10^{-5} $ K^s - K^d = 3.6 \times 10^{-6}$	a policy that is not optimal an early stop

- **Five checks, each aimed at a different piece. Without them there is nothing to grade a network against.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A.

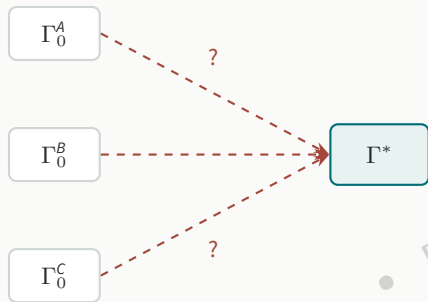
Session 4.1

What the Classical Solve Cannot Answer

One fixed point is not a law of motion, and the distribution is now a state



One Fixed Point, Not a Law of Motion



each dashed path is a separate problem: a sequence of prices, solved backward and forward

- ▶ Stage 1A says where the economy rests, not how it gets there.
- ▶ A transition from one Γ_0 is a price path and a shooting loop (Deck 4.3). A second Γ_0 is a second loop.
- ▶ Welfare along the way, or a policy that reacts to Γ , needs the answer at every Γ .
- ▶ **The unknown changes from a number, r^* , to a function of a distribution: $g(\varepsilon, a; \Gamma)$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Deck 4.3, *The Unknown Is a Path of Prices*.

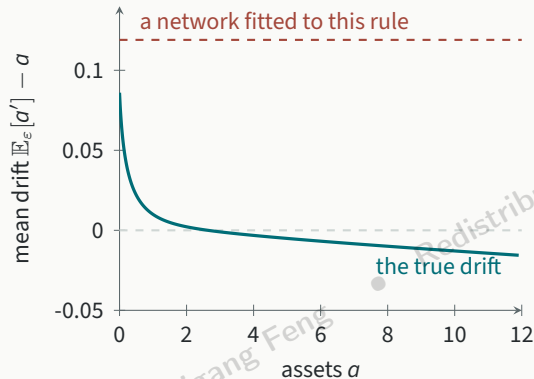
The Distribution Is a State

Route	What the network reads	Its weakness here
a simulated panel	every household's (ε, a) , in some order	noise, and an ordering to forget
generalised moments	learned averages of the panel	the moments must be chosen or learned
a histogram	the masses in N fixed bins	N inputs; the grid must be chosen

- ▶ Each route turns an infinite-dimensional Γ into a finite vector. Deck 4.2 takes the first two; here, $N = 140$ bins.
- ▶ **A histogram has no ordering to forget, and it can be moved forward exactly, by a matrix.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar et al. (2021); Azinovic et al. (2022); Han, Yang & E (2026).

A Near Unit Root Makes Small Errors Large



- ▶ The true rule's mean drift stays within 0.022 of zero for $0.5 \leq a \leq 10$.
- ▶ With $\beta(1+r) = 0.99763$, a one-signed bias in it is amplified about $1/(1 - \beta R) \approx 420$ times.
- ▶ Measured: a network outputting the level of a' , fitted to the exact rule, kept a drift of +0.119; its K was 15.3, not 3.6595.
- ▶ **So the network outputs a change in assets.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A and the network initialisation.

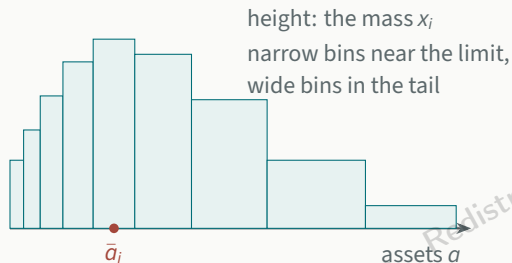
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Characterising the Evolution of Γ

A histogram, a quadrature, and a transition matrix rebuilt inside the computational graph



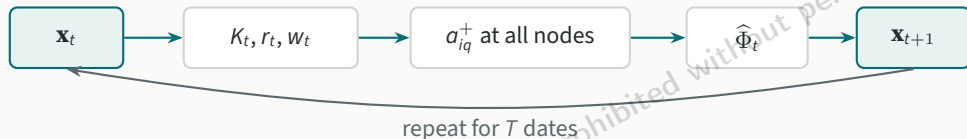
A Histogram and a Quadrature



- ▶ $N = 140$ bins on $[0, 16]$ with midpoints $\bar{a}_i$; the state is $\mathbf{x}$, $\sum_i x_i = 1$, $K = \sum_i x_i \bar{a}_i$.
- ▶ ε : $Q = 5$ Gauss-Hermite nodes ε_q , weights p_q . An expectation is a five-term sum.
- ▶ Grid chosen once from the benchmark: no bin above 1.6% of households, nothing off the top.
- ▶ **The discretisation is part of the model. Fix it for every method compared.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

The Law of Motion, Rebuilt at Every Date



From one distribution $\mathbf{x}_0$, for $t = 0, \dots, T-1$

- 1 $K_t = \sum_i x_{t,i} \bar{a}_i$; r_t, w_t from the firm, hours in closed form: no price loop
- 2 saving $a_{iq}^+ = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$ at every node; consumption from the budget
- 3 build the $N \times N$ matrix $\hat{\Phi}_t$ and set $\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}_t$

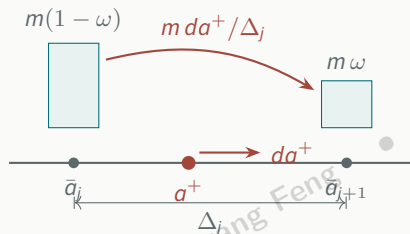
► $\hat{\Phi}_t$ depends on $\mathbf{x}_t$ and on the network, so it is rebuilt at every date of every rollout: computed, never simulated, inside the graph.

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Building $\hat{\Phi}$, and Why This Kernel

For every source midpoint $\bar{a}_i$ and node ε_q

- 1 evaluate the saving network once: $a^+ = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x})$
- 2 find the bracket $\bar{a}_j \leq a^+ \leq \bar{a}_{j+1}$, and the weight $\omega = (a^+ - \bar{a}_j) / (\bar{a}_{j+1} - \bar{a}_j)$
- 3 add $p_q(1 - \omega)$ to $\hat{\Phi}_{ij}$ and $p_q \omega$ to $\hat{\Phi}_{i,j+1}$



- **Exact in the mean:** the only split between neighbours with mean a^+ .
- **Mass by construction:** each row of $\hat{\Phi}$ sums to one.
- **Differentiable:** da^+ moves mass $m da^+ / \Delta_j$ between neighbours, and autodiff chains it across dates.
- **Stationary K within 0.06% of the exact operator's; an inversion in ε was off by 17%.**

Session 4.1

Stage 1B: An Actor–Critic Solve

The saving rule is the actor, the household's value is the critic, and both read the histogram



The Problem the Two Networks Solve

- ▶ At a distribution $\mathbf{x}$, each household solves, taking $\mathbf{x}$, prices and $\mathbf{x}_+$ as given,

$$V(\varepsilon, a; \mathbf{x}) = \max_{0 \leq a' \leq \bar{a}} \left\{ u\left(c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}\right) + \beta \sum_{q'} p_{q'} V(\varepsilon_{q'}, a'; \mathbf{x}_+) \right\}, \quad u(y) = \frac{y^{1-\sigma}}{1-\sigma},$$

with $c = (1 + r(\mathbf{x})) a + w(\mathbf{x}) \varepsilon l(\varepsilon; \mathbf{x}) - a'$ and $\mathbf{x}_+ = \mathbf{x} \hat{\Phi}(\mathbf{x}; \theta)$.

- ▶ **Actor:** $a' = g_\theta(\varepsilon, a; \mathbf{x}) = \min\{\max\{\tilde{a}, 0\}, \bar{a}\}$, $\tilde{a} = a + D(2s_\theta - 1)$, $s_\theta \in (0, 1)$, $D = 1.3$.
- ▶ **Critic:** $V_\psi(\varepsilon, a; \mathbf{x}) \approx V$, and a slowly moving copy $V_{\psi-}$, the **target network**.
- ▶ Each: two hidden layers of 256 rectified-linear units, reading (ε, a) and the 140 masses.
- ▶ **No government, so no one else's problem: prices are marginal products at $\mathbf{x}$, and the Euler equation is the household's own first-order condition.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

The Critic Step: Fit the Bellman Equation

- ▶ For a batch of distributions $\{\mathbf{x}^e\}$ and every node $(\bar{a}_i, \varepsilon_q)$, with $a_{iq}^+ = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}^e)$ from the current actor:

$$\min_{\psi} \mathcal{L}_V(\psi) = \text{mean}_{e,i,q} \left[V_{\psi}(\varepsilon_q, \bar{a}_i; \mathbf{x}^e) - y_{iq}^e \right]^2, \quad y_{iq}^e = U_{iq}^e + \beta \sum_{q'} p_{q'} V_{\psi^-}(\varepsilon_{q'}, a_{iq}^+; \mathbf{x}_+^e).$$

- ▶ U_{iq}^e is flow utility at a_{iq}^+ . The target y is held fixed while ψ moves: no gradient flows into it.
- ▶ One Adam step, learning rate 3×10^{-3} ; then $\psi^- \leftarrow (1 - 0.02) \psi^- + 0.02 \psi$.
- ▶ The expectation over ε' is an exact five-node sum: the target carries no sampling noise.
- ▶ **8 such steps per iteration; before training starts, 6000 with the actor frozen.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

The Actor Step: Raise the Right-Hand Side

- ▶ With the critic fixed, the actor maximises the Bellman right-hand side at every node:

$$\max_{\theta} J(\theta) = \text{mean}_{e,i,q} \left[U(a_{iq}^+(\theta)) + \beta C_{iq}(a_{iq}^+(\theta)) - \kappa B_{iq}(\theta) \right],$$

where $C_{iq}(a') = \sum_{q'} p_{q'} V_{\psi^-}(\varepsilon_{q'}, a'; \mathbf{x}_+^e)$ is the continuation and $\kappa = 1$.

- ▶ Barrier $B = [\max(-c, 0) + \max(-c + \psi_l \frac{l^1 + \theta_l}{1 + \theta_l}, 0)] / \eta_c$, $\eta_c = 1 \times 10^{-4}$: zero where consumption is feasible.
- ▶ The slope of C is replaced: its *value* comes from the critic, its *derivative* from the envelope condition, $\partial C / \partial a' \equiv \sum_{q'} p_{q'} (1 + r(\mathbf{x}_+)) u'(c_{+,q'})$, with c_+ from the current actor at $(\varepsilon_{q'}, a'; \mathbf{x}_+)$.
- ▶ So $\partial J / \partial a^+ = -u'(c) + \beta \mathbb{E}[(1 + r_+) u'(c_+)]$, the Euler residual.
- ▶ **Why: at $\beta R = 0.99763$ the two terms nearly cancel; a fitted critic's slope is too noisy to resolve it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

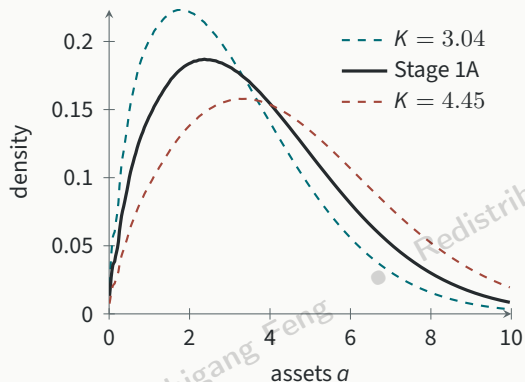
The Actor's Full Loss: One Economic Term, Two Solver Terms

$$\mathcal{L}_g(\theta) = -J(\theta) + \lambda_K \frac{1}{T} \sum_{t=1}^T (K_t(\theta) - K^*)^2 + \lambda_E \text{mean}_e \sum_{i,q} x_i^e p_q |\text{FB}(\vartheta_{1,iq}, \vartheta_{2,iq})|$$

- **Rollout:** $\mathbf{x}_0 = \mathbf{x}^*$, $\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}(\mathbf{x}_t; \theta)$, $K_t = \sum_i x_{t,i} \bar{a}_i$; $T = 100$, $\lambda_K = 1000$. The gradient passes through all T matrices. The target is Stage 1A's K^* : **agreement on K is imposed.**
- **Euler with the limit:** $\text{FB}(\vartheta_1, \vartheta_2) = \vartheta_1 + \vartheta_2 - \sqrt{\vartheta_1^2 + \vartheta_2^2 + \eta^2}$, with $\vartheta_1 = [u'(c) - \beta \mathbb{E}(1 + r_+) u'(c_+)] / u'(c)$ and $\vartheta_2 = \tilde{a} / 0.1$; $\eta = 0.01$, $\lambda_E = 10$.
- $\text{FB} = 0$ exactly when $\vartheta_1 \geq 0$, $\vartheta_2 \geq 0$ and one is zero: the Euler equation holds, or the limit binds. Weighted by where households are.
- **2 Adam steps per iteration, rate 5×10^{-5} . At βR near one, the rollout, not a residual, pins K .**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B; Maliar, Maliar & Winant (2021).

Training Data: Whole Distributions



- ▶ A training example is a whole distribution $\mathbf{x}^e$, never a household.
- ▶ $x_i^e \propto x_i^* e^{\kappa \bar{a}_i}$, κ uniform on ± 0.14 : a tilt of the benchmark.
- ▶ K spans 3.04 to 4.45. The band sets where losses are evaluated, nothing else.
- ▶ **At each $\mathbf{x}^e$ every node $(\bar{a}_i, \varepsilon_q)$ is evaluated, weighted $x_i p_q$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

The Algorithm

Stage 1B: the household's actor-critic, no government

- 1 initialise the actor by regression on EGM rules at constant prices, one per sampled K
 - 2 pre-fit the critic to that actor, 6000 steps, so the actor cannot exploit a bad critic
 - 3 draw 128 distributions from the tilted band
 - 4 8 critic steps on $\mathcal{L}_V$, each followed by $\psi^- \leftarrow (1 - 0.02)\psi^- + 0.02\psi$
 - 5 2 actor steps on $\mathcal{L}_g$: envelope gradient, rollout penalty, Euler term
 - 6 repeat 4000 times; at intervals, find the actor's own stationary $\mathbf{x}$ and score it
- ▶ Step 1 means the training loop starts at a correct answer. The test is whether it keeps it while learning how the rule depends on $\mathbf{x}$.
 - ▶ **The final run: about 2.9 GPU-hours, against 30 CPU-seconds for Stage 1A.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

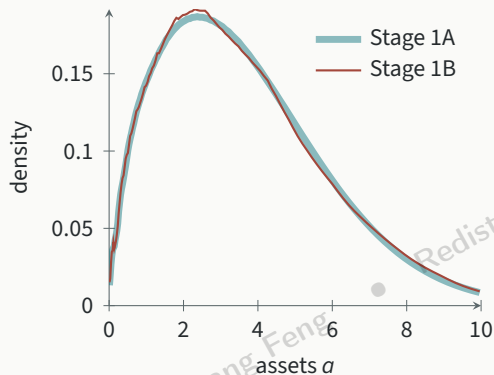
Session 4.1

What It Returns, and Where It Stops

The network against the benchmark, the distribution in motion, and why persistence changes the problem



Stage 1B Against Stage 1A

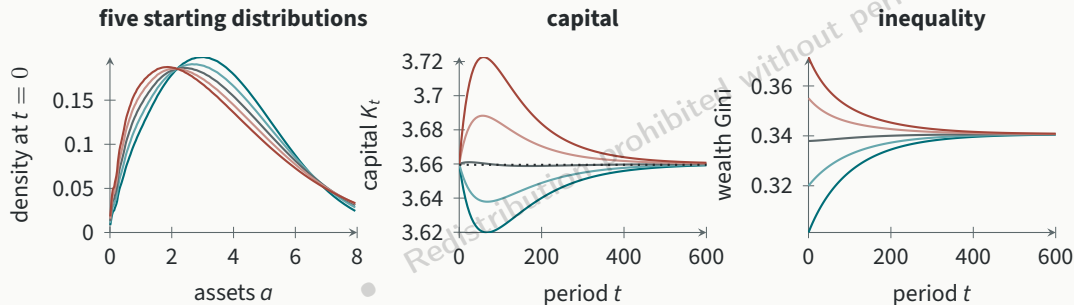


	1A	1B	gap
capital K	3.6595	3.6600	0.01%
wealth Gini	0.3381	0.3407	0.8%
mass below $a = 0.6$	0.0415	0.0419	0.9%
L_1 distance			0.024

- K is held by the penalty. The Gini, the mass near the limit and the whole density are not targeted, and agree to under 1%.
- Euler error, mass-weighted: 0.16%; over all nodes, 0.48%.

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. The 1A density is integrated onto the network's 140 bins.

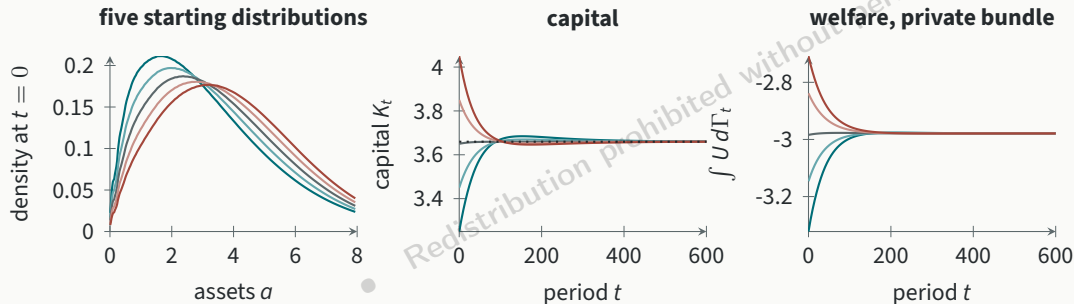
Same Capital, Different Inequality



- ▶ All start at $K_0 = 3.660$, Gini 0.301–0.372; all reach Gini 0.341, from below and above.
- ▶ **K leaves and returns (3.620–3.723). A state summarised by its mean would see nothing.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^* .

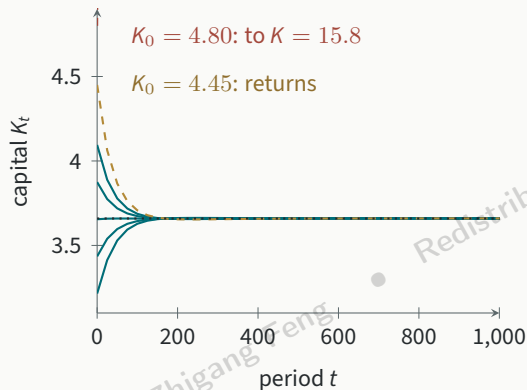
Same Spread, Different Capital



- ▶ K_0 from 3.25 to 4.05: K returns in about a hundred periods, the Gini far later.
- ▶ **Welfare along the path, from any start: what Stage 1A cannot return.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^* . No government, so no public good.

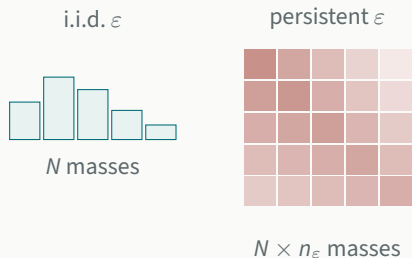
One Attractor, and the Edge of the Band



- ▶ 5 starts with K_0 from 3.21 to 4.09: one Γ^* , terminal K within 3.4×10^{-8} .
- ▶ Within 10^{-3} after 50–475 periods, 10^{-6} after 1125–1325: βR near one makes it slow.
- ▶ Outside the band, $K_0 = 4.45$ returns; $K_0 = 4.80$ runs to the top of the grid.
- ▶ **The rule is valid where it was trained.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B; dotted: Stage 1A's K^* .

Where the Characterisation Gets Hard



a and ε become correlated: the poor are poor because their ε was low

- Persistent ε : $\Gamma(\varepsilon, a)$ no longer factorises. The state is a joint histogram, and $\hat{\Phi}$ must carry the chain Π .
- With i.i.d. risk only 0.009% sit at the limit. In the lab's economy, with persistence 0.9, 2.9% do: runs of low ε walk them down to it.
- An aggregate shock z makes the map $\Phi(\Gamma, z, z')$ random, to be learned on every branch.
- **Each step multiplies the size of $\hat{\Phi}$ and of every rollout in memory.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the lab, block A; Krusell & Smith, *JPE* (1998).

Bridge: When Φ Cannot Be Written Down, Simulate It

This deck. Γ is a histogram, Φ a matrix computed in the graph: exact, deterministic, differentiable.

persistence, aggregate risk

Next. Γ is a simulated panel, and the network learns from the states that the panel visits.

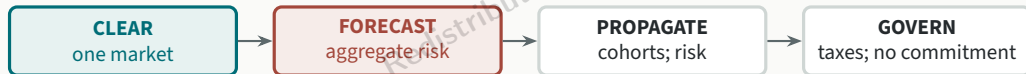
- ▶ Deep Equilibrium Nets and DeepHAM simulate households, and train where the simulation goes.
- ▶ The price: sampling noise, no derivative through a head count, coverage only where the policy goes.
- ▶ The gain: persistence and aggregate risk cost a longer simulation, not a bigger matrix.
- ▶ **Deck 4.2: three such methods. Deck 4.4: this histogram, with a government.**

Source: Azinovic, Gaegauf & Scheidegger, *IER* (2022); Han, Yang & E (2026).

Session 4.2

Krusell-Smith, Solved Three Ways

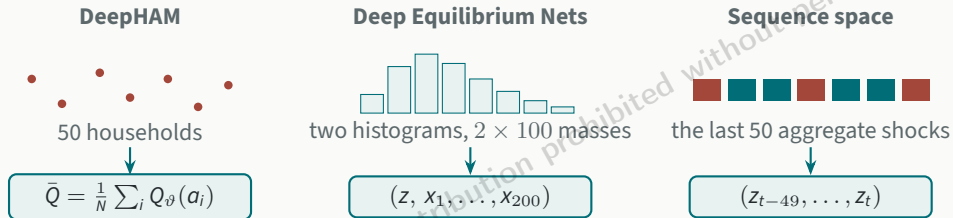
Generalised moments, a histogram, a history of shocks: three ways to put a moving distribution into a network, and what each can and cannot see



Zhigang Feng

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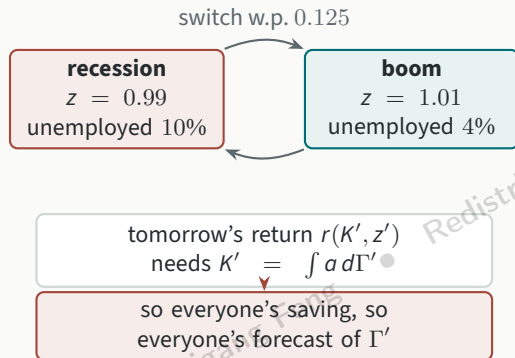
Three Ways to Give a Network a Moving Distribution



- Krusell and Smith's economy: the household's state is $(a, \varepsilon; z, \Gamma)$, and Γ is a distribution.
- Each replaces Γ by something finite. For each: what the network reads, what it minimises, *where its training distributions come from*.
- **Then Deck 4.1's method, which writes down how Γ moves.**

Source: Han, Yang & E, *QE* (2026); Azinovic, Gaegauf & Scheidegger, *IER* (2022); Azinovic-Yang & Žemlička (2025).

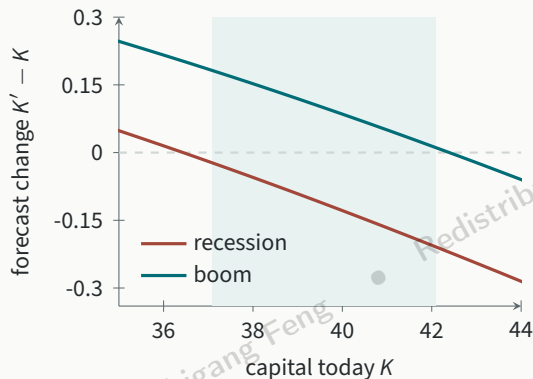
The Economy, and Why the Household Must Forecast Γ



- ▶ $V(a, \varepsilon; z, \Gamma) = \max_{a' \geq 0} \ln c + \beta \mathbb{E} V(a', \varepsilon'; z', \Gamma')$,
 $c + a' = (1 - \delta + r)a + y(\varepsilon)$.
- ▶ r, w are marginal products at $K = \int a d\Gamma$; employment $\varepsilon \in \{0, 1\}$ moves with z .
- ▶ $\Gamma' = H(\Gamma, z, z')$: **the law of motion every household needs and nobody is given.**

Source: Krusell & Smith, *JPE* (1998); Den Haan, Judd & Juillard, *JEDC* (2010).

The Classical Answer: Forecast With the Mean



- ▶ The bet: $\log K' = \kappa_0(z) + \kappa_1(z) \log K$. The state shrinks to (a, ε, K, z) .
- ▶ Solve at the rule, push a histogram 3,000 quarters, regress, update: 17 CPU-seconds.
- ▶ $R^2 > 0.99999$ one step ahead. Run alone for 2,000 quarters: mean miss 0.054%, worst 0.187%.
- ▶ **The benchmark, and the idea each deep method generalises.**

Source: Krusell & Smith (1998); Den Haan, *JEDC* (2010); the lab, block B. Quarterly, as published: $\alpha = 0.36, \beta = 0.99, \delta = 0.025, u = \ln c, z \in \{0.99, 1.01\}$.

Four Questions Every Deep Method Answers

	DeepHAM	Deep Equilibrium Nets	Sequence space
1. what stands for Γ	learned moments of a 50-household panel	a 100-point histogram per employment state	the last 50 aggregate shocks
2. what is minimised	minus lifetime utility over a rollout; a value regression	Euler and KKT residuals	a Fischer–Burmeister Euler residual
3. where training states come from	simulated panels, refreshed	16 simulated economies, carried forward	a replay buffer of simulated histories
4. what the gradient passes through	150 simulated periods of the panel	one period, at stored next states	one period; prices held fixed

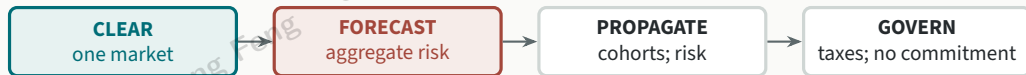
- **Row 3: in all three, training distributions come from simulating the network's own policy.**

Source: DeepHAM's public repository; the Deep Equilibrium Nets continuum-of-agents code; the sequence-space tutorial.

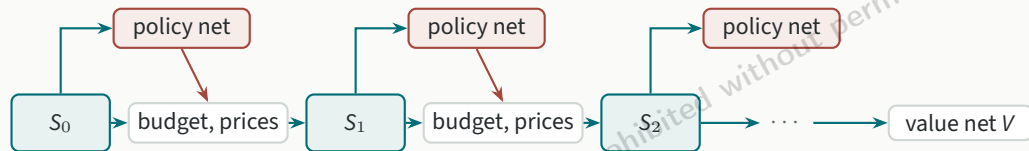
Session 4.2

Method 1: DeepHAM

A network inside an average learns the moments that matter, and the policy is trained on simulated paths



DeepHAM in One Picture: an Unrolled Panel

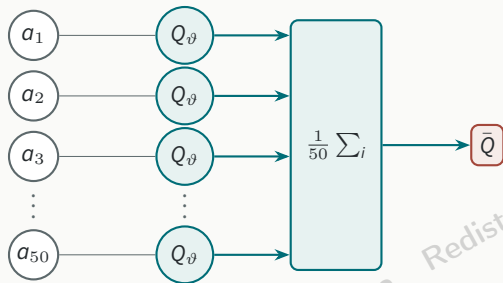


S_t : all 50 households' (a_i, ε_i) and z_t ; prices from $K_t = \frac{1}{50} \sum_i a_i$

- ▶ 384 economies run side by side, each a panel of $N = 50$ households.
- ▶ The actor simulates 150 periods forward from a stored cross-section inside the loss; a critic values what is left. Where the stored cross-sections come from: the training-loop pages.
- ▶ **Both networks read the panel through generalised moments.**

Source: Method 1 follows Han, Yang & E, "DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks", *QE* 17(2), 2026 (preprint arXiv 2112.14377), and its public repository (Figure 2, §3).

Generalised Moments: a Network Inside an Average

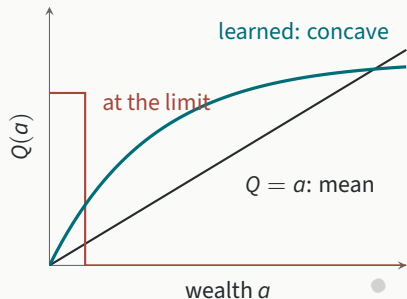


one network, shared by all:
2 layers of 12 tanh units

- ▶ $\bar{Q}_t = \frac{1}{N} \sum_i Q_{\vartheta}(a_t^i)$: one small network, applied to each household's wealth, then averaged.
- ▶ Household i 's policy reads $(a_i, \varepsilon_i, z_t, \bar{Q}_t)$.
- ▶ ϑ is trained with the policy and value networks: the moment is whatever lowers the loss.
- ▶ **Krusell-Smith run: $J = 1$, one generalised moment of wealth.**

Source: The paper, §2.2; repository *util.py* and its Krusell-Smith configuration.

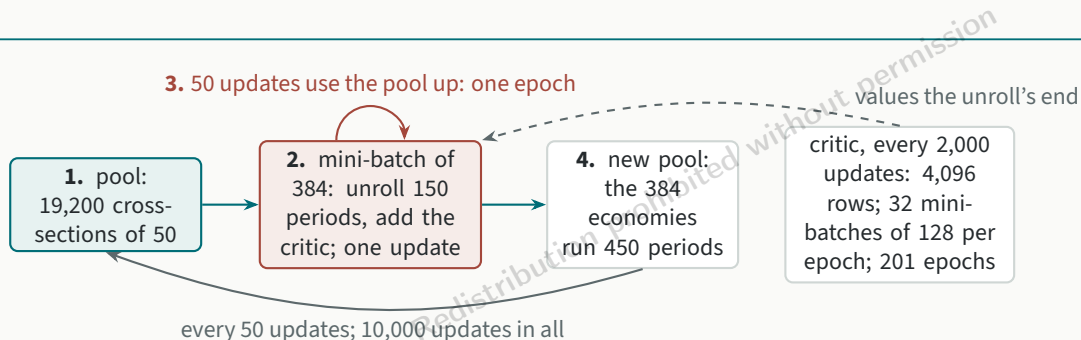
Why an Average, and What It Nests



- ▶ **Permutation invariance:** renumbering households leaves $\sum_i Q(a_i)$ unchanged.
- ▶ $Q(a) = a$ gives the mean, Krusell and Smith's state; an indicator of the limit gives the hand-to-mouth share.
- ▶ Learned here: Q concave in wealth, value linear in $\bar{Q}$. Moving wealth from rich to poor raises $\bar{Q}$ at the same K .
- ▶ **Bellman error: the Krusell–Smith rule 0.0253; DeepHAM with the mean 0.0184; with one generalised moment 0.0151.**

Source: The paper, §3.2, Table 2, Figure 3. Curves schematic.

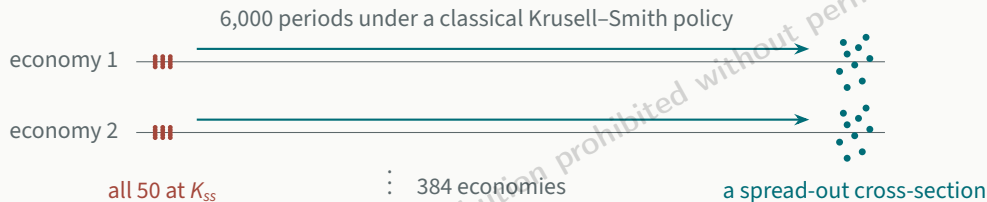
DeepHAM's Training Loop in One Picture



- ▶ **1. Samples:** a pool of 19,200 cross-sections of 50, from the 384 burned-in economies.
- ▶ **2.** A mini-batch is 384 of them.
- ▶ **3–4.** 50 updates use the pool up; then a new pool. The critic refits in its own loop.
- ▶ **The 150-period unroll is inside the loss: it moves nothing in the pool.**

Source: Repository *train_KS.py*, *policy.py* (*train*, *sampler*), *dataset.py*; the Krusell–Smith configuration.

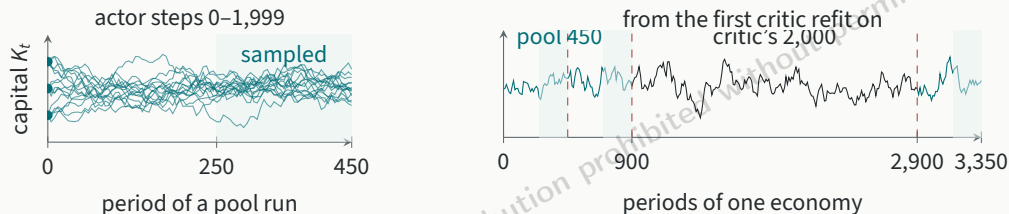
1. Samples: 384 Economies of 50, Burned In



- ▶ 384 separate economies run side by side, each with 50 households and its own aggregate shocks.
- ▶ Each starts with everyone at the representative-agent capital; employment drawn at the aggregate state's unemployment rate.
- ▶ After 6,000 periods the 384 final cross-sections are the **initial states**.
- ▶ **A “distribution” is a sample of 50 households; the algorithm holds 384 of them.**

Source: Repository *dataset.py* (*KSInitDataSet*), *simulation_KS.py*, the Krusell–Smith configuration.

1. Samples Over Training: 384 Tracks



- ▶ Left, actor steps 0-1,999: every pool re-run starts from the *same* burned-in states (6,000 periods from K_{ss}), on fresh shocks.
- ▶ Right, from the first critic refit: each re-run continues where the last ended; about 80,000 periods per economy by step 10,000.
- ▶ **Like Method 2: long simulations under the evolving policy; each datum is 50 households.**

Source: Repository `dataset.py` (`update_from_simul`), `policy.py` (`train`, `sampler`), the Krusell-Smith configuration (`update_init`). Paths schematic.

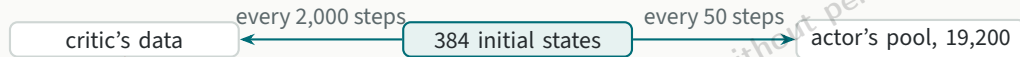
2-3. Actor: Mini-Batches of 384; the Pool Used Up in 50 Steps



- ▶ **Dataset:** the pool, 19,200 cross-sections from a 450-period run of the 384 economies.
- ▶ **Mini-batch:** the pool is shuffled once; each update takes the next 384 cross-sections as the starting states S_0 of its unroll.
- ▶ **Inside the loss:** from each S_0 , fresh shocks and 150 simulated periods; household 1's discounted utility is one sample. This does not move the pool.
- ▶ **Epoch:** $19,200/384 = 50$ updates use every snapshot once; then the pool is rebuilt, the next episode.

Source: Repository `dataset.py` (`get_policydataset`, `crazyshuffle`) and `policy.py` (`sampler`, `train`).

4. Episodes: Three Clocks, and the Algorithm



after the first refit, every re-simulation moves the initial states

DeepHAM, Krusell–Smith configuration

- 1 burn in 384 economies of 50 for 6,000 periods; fit critics under a constant consumption share
- 2 build the pool: 450 periods, 50 snapshots per economy
- 3 10,000 actor steps: 384 cross-sections from the pool, unroll 150, one Adam step
- 4 every 50 steps: re-simulate the pool; every 2,000: re-simulate the critic's data and refit

► **Every training distribution is a stretch of these 384 simulated economies, under the evolving policy.**

Source: Repository *train_KS.py*, *policy.py* (*train*, *sampler*), *dataset.py* (*update_from_simul*).

The Critic's Own Loop: Data, Mini-Batches, Epochs



- **Data:** 4,608 cross-sections of 50 (230,400 targets); 512 set aside to check the fit, 4,096 for training.
- **Rows:** each of the 384 economies runs 2,000 periods from its initial state under the current actor; its cross-sections at the 12 start dates are the rows.
- **Samples:** each household's target is its own realised 800-period discounted utility along that same path.
- **Mini-batches and epochs:** 128 shuffled rows per Adam step, so $4,096/128 = 32$ per epoch; 201 epochs; three critics. Rebuilt every 2,000 actor updates.

Source: Repository *dataset.py* (*get_valuedataset*, *process_vdatadict*), *value.py*; the Krusell–Smith configuration.

5a. Inside One Actor Update: the Loss

- **Unroll**: simulate 150 periods forward *inside the loss*, from each S_0 in the batch B , so the gradient runs back through every period:

$$\max_{\theta} \frac{1}{384} \sum_{S_0 \in B} \left[\sum_{t=0}^{148} \beta^t \ln c_t^1 + \beta^{149} \bar{V}(S_{149}) \right], \quad c_t^i = \sigma(\mathcal{C}_{\theta}(a_i, \varepsilon_i, z_t, \bar{Q}_t)) \cdot \text{wealth}_t^i.$$

- $\bar{V}$ is the average of the three critics; $\beta^{149} \approx 0.22$ leaves the rest of the future to them. Adam, rate 4×10^{-4} .
- Only household 1 optimises; the other 49 use the same network with the gradient stopped, and set prices and $\bar{Q}$: **fictitious play**.
- **Not 384 cross-sections $\times$ 50 households per step: the reshuffle only makes household 1 a random member. Averaging all 50 would be a planner's problem.**

Source: The paper, eq. (10), Algorithm 1; repository *policy.py* (*loss*, the “socialplanner” option).

5b. Inside One Critic Update: the Loss

- ▶ A mini-batch B is 128 training cross-sections (s, e) ; every household in each counts:

$$\mathcal{L}(\psi; B) = \frac{1}{128 \cdot 50} \sum_{(s,e) \in B} \sum_{i=1}^{50} \left[v_{\psi}(a_{i,s}^e, \varepsilon_{i,s}^e, z_s^e, \bar{Q}_s^{V,e}) - \hat{V}_{i,s}^e \right]^2, \quad \hat{V}_{i,s}^e = \sum_{\tau=0}^{799} \beta^{\tau} \ln c_{i,s+\tau}^e.$$

- ▶ One Adam step, rate 10^{-4} , per mini-batch: 32 per pass, about 6,400 per critic.
- ▶ A row keeps its 50 households together: each household's input includes $\bar{Q}_s^{V,e}$, the moment of its own cross-section.
- ▶ $\beta^{800} < 3.3 \times 10^{-4}$: the truncated sum stands in for the infinite one.
- ▶ **Monte Carlo evaluation, as in Session 3: each target is one realised future, and the regression averages the noise.**

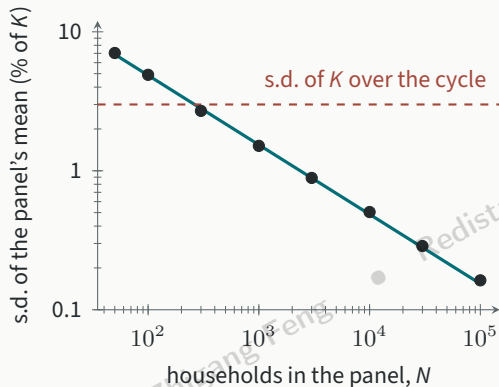
Source: The paper, eq. (8)–(9); repository *dataset.py*, *value.py*.

Ten Numbers, and What Each One Is

number	what it counts	why
50	households in one economy	the finite panel that stands for a continuum
384	economies side by side; the actor's batch	one draw of Γ each
6,000	periods of burn-in, once	from everyone at K_{ss} to a spread-out sample
2,000	periods simulated for the critic	12 start dates plus 800 periods after the last
100 / 800	spacing of the critic's start dates / length of each future	snapshots apart; β^{800} negligible
450 / 250 / 4	periods simulated for the actor / discarded / spacing of snapshots	adjust to the policy, then sample
19,200	cross-sections in the actor's pool	384 economies $\times$ 50 snapshots
150	periods in one actor rollout	$\beta^{149} \approx 0.22$ leaves the rest to the critic
128 / 6,400	critic mini-batch: cross-sections / household targets	one Adam step each; 32 per epoch
50 / 2,000	actor steps per pool / per critic refit	how often each dataset is rebuilt

Source: Repository Krusell–Smith configuration, *dataset.py*, *policy.py*.

What 50 Households Cost



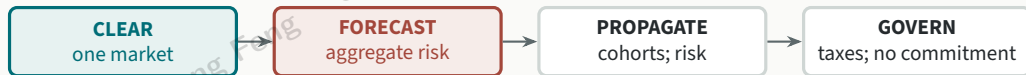
- ▶ Wealth has s.d. 18.9 around $K = 38.8$: a panel's mean misses K by $18.9/\sqrt{N}$.
- ▶ $N = 50$: 2.67, 2.3 times the s.d. of K over the whole cycle.
- ▶ The paper treats $N = 50$ as the model, with similar results at 100 and 200.
- ▶ **A sampled state, and prices that inherit its noise.**

Source: Lab block B, 400 panels per N . Quarterly, as published: $\alpha = 0.36, \beta = 0.99, \delta = 0.025, u = \ln c, z \in \{0.99, 1.01\}$. The paper, fn. 2.

Session 4.2

Method 2: Deep Equilibrium Nets

Keep the whole histogram, and ask only that the first-order conditions hold one period ahead



The Same Economy, With a Histogram as the State

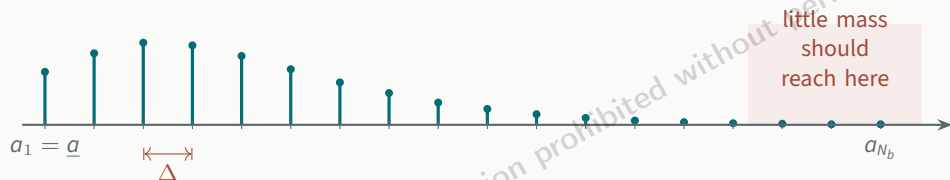


one asset grid of 100 points;
a histogram of the unemployed
and one of the employed

- ▶ State $X = (z, h_u, h_e)$: the shock, and two histograms on one asset grid: 2×100 masses.
- ▶ Two, because employment is persistent: who is poor today predicts who is unemployed tomorrow, so assets alone do not determine next period.
- ▶ One network: $(\varepsilon, a; X) \mapsto (a', \lambda)$. Prices follow from K .
- ▶ **Unlike DeepHAM: every mass is an input; no learned moment, no critics.**

Source: Method 2 follows Azinovic, Gaegauf & Scheidegger, “Deep Equilibrium Nets”, *IER* 63(4), 2022, and its public continuum-of-agents code, adapted to Krusell–Smith. Here: App. A.5; the code’s state. A mean-capital version: lab block D.

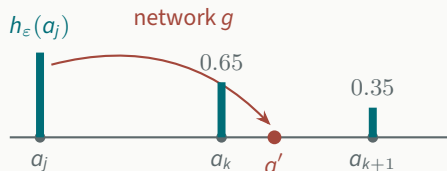
Encoding the Distribution: Mass on a Grid of Points



- ▶ N_b equidistant points: the first is the limit $\underline{a}$; the last high enough that little mass reaches it. Savings above it are set to it.
- ▶ The code: 100 points on $[0, 20]$, $\Delta \approx 0.2$. A mass $h(a_j)$ sits on each point: a finite vector the network can read, and move exactly.
- ▶ **Unlike DeepHAM: the distribution is written on a grid, not sampled by 50 households.**

Source: The paper, following Reiter (2009) and Young (2010); the continuum-of-agents code (the asset grid).

From Today's Histogram to Tomorrow's: One Lottery Step



For each ϵ and grid point a_j

- 1 saving $a' = g(\epsilon, a_j; X)$, set to a_{N_b} if above
- 2 a' lies between a_k and a_{k+1}
- 3 weights $w_k = 1 - |a_k - a'|/\Delta$, $w_{k+1} = 1 - w_k$
- 4 send $w_k h_\epsilon(a_j)$ to a_k , $w_{k+1} h_\epsilon(a_j)$ to a_{k+1}

$$h'_{\epsilon'}(a_k) = \sum_{\epsilon} \pi(\epsilon' | \epsilon; z, z') \sum_j h_\epsilon(a_j) w_k(a'_{\epsilon j})$$

- ▶ Then employment moves between the two histograms, with $\pi(\epsilon' | \epsilon; z, z')$: one h' for each z' . Mass is conserved; the mean of a' is exact.
- ▶ **Unlike DeepHAM: tomorrow's distribution is computed, 200 network calls, not simulated household by household.**

Source: The continuum-of-agents code (the grid weights and the next-period histogram); Young, *JEDC* (2010).

The Equilibrium Conditions Are the Loss

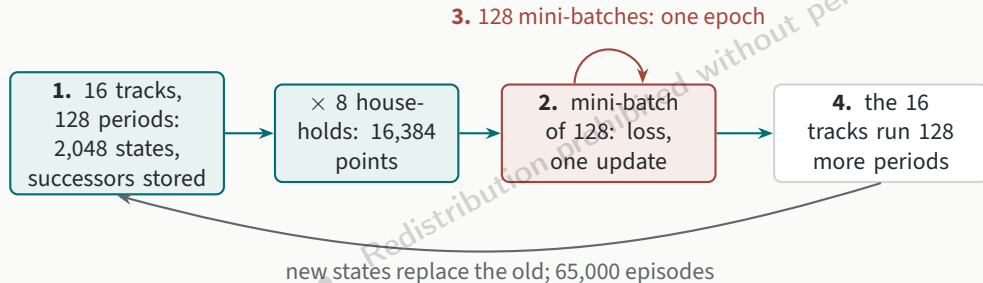
- ▶ At a point $(\varepsilon, a; X)$, with $c = (1 - \delta + r)a + y(\varepsilon) - a'$ and c' from the same network at $(\varepsilon', a'; X')$:

$$R_{EE} = \frac{\left[\beta \mathbb{E}[(1 - \delta + r')/c'] + \lambda \right]^{-1}}{c} - 1, \quad R_{KKT} = a' \lambda.$$

- ▶ $\mathbb{E}$: an exact sum over the four (z', ε') . $R_{EE} = 0$ is the Euler equation; $R_{KKT} = 0$ lets $\lambda > 0$ only at the limit.
- ▶ $X' = (z', h'_u, h'_e)$ is the lottery step of the last page, one for each z' .
- ▶ **Unlike DeepHAM: first-order conditions one period ahead, not utility over a 150-period rollout.**

Source: The paper; the continuum-of-agents code (its Euler and KKT terms).

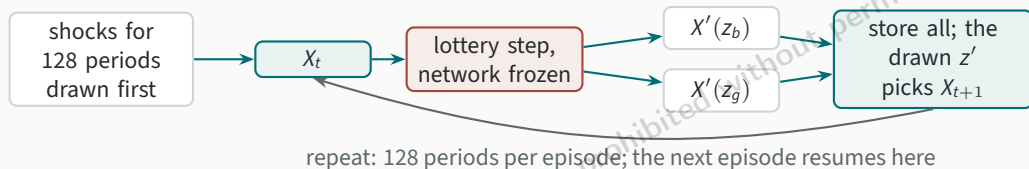
Deep Equilibrium Nets' Training Loop in One Picture



- ▶ **1. Samples:** 16 tracks $\times$ 128 periods, successors stored; 8 households per state.
- ▶ **2–4.** Shuffled mini-batches of 128; one epoch; then the tracks continue.
- ▶ **No look-ahead in the loss: the successors were stored in the simulation.**

Source: The continuum-of-agents code (the episode loop, training data, batching).

1. Samples: Simulated, and Stored With Their Successors



- ▶ **Simulation for data:** 16 tracks, 128 periods each per episode; the next page shows where they start and how they continue.
- ▶ Each period: one lottery step per z' ; all successors stored; the pre-drawn shock picks the next state.
- ▶ **Inside the loss:** no simulation; $t + 1$ is a stored successor.
- ▶ **A one-period condition at states from a long simulation; DeepHAM simulates 150 periods inside its loss.**

Source: The continuum-of-agents code (shock simulation, the forward pass, the episode loop).

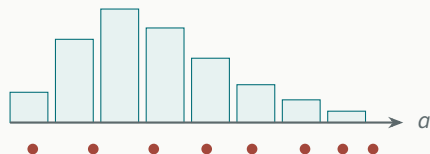
1. Samples: 16 Tracks, 128 Periods an Episode



- ▶ Episode 1: 16 copies of *one* fixed state (wealth near zero), differing only by their shock draws.
- ▶ Each episode: 128 periods per track, network frozen, $16 \times 128 = 2,048$ states; then train. At each dashed line, every track continues from its own end state, under the updated network.
- ▶ **Not 16 hand-picked starts: 16 long simulations, in 128-period pieces.**

Source: The continuum-of-agents code (starting states, shock simulation, the episode loop). The paths are schematic.

1. Samples: Each State Paired With 8 Households



- **Data:** the $16 \times 128 = 2,048$ aggregate states just simulated, each with its stored successors.
 - **Evaluation points:** each state X is paired with 8 household states (ε, a) , drawn at random: a uniform on $[0, 20]$, ε a fair coin. 16,384 points $(\varepsilon, a; X)$.
- X is not a starting point: nothing is simulated from it. It is where the loss is evaluated, with its stored successors $X'(z')$.
 - Why points: the network is pointwise, $(\varepsilon, a; X) \mapsto (a', \lambda)$, and the Euler condition is a household's. Why uniform: the lottery uses the policy at all 200 grid points.

Source: The continuum-of-agents code (training-data construction).

2-5. Mini-Batches, One Epoch, and the Loss

- ▶ **Mini-batches**: the 16,384 points are shuffled and cut into 128 mini-batches B of 128. **Inside one update**, each point gives one Euler and one KKT residual:

$$\mathcal{L}(\theta; B) = \frac{1}{128} \sum_{(\epsilon, a; X) \in B} \left[R_{EE}^2 + R_{KKT}^2 \right], \quad \text{one Adam step at rate } 10^{-5}.$$

- ▶ The successors X' were stored during simulation: the network is re-evaluated at them, but how the histogram moves is not differentiated.
- ▶ **Epoch**: the 128 mini-batches use every point once; then the next episode, the 16 tracks run 128 more periods.
- ▶ One period ahead only: the long simulation chooses the states; the loss never looks past $t + 1$.
- ▶ **Unlike DeepHAM: no rollout inside the loss and no critic; the gradient sees one period, at stored states.**

Source: The continuum-of-agents code (the loss and the training step).

Deep Equilibrium Nets: the Numbers, and What Each One Is

number	what it counts	where it appears
100 / 200	grid points / histogram masses (2×100)	the state X , with the shock
16	tracks (economies), identical at the start	each continues from its own end state
128	periods per economy per episode	$2,048/16$
2,048	aggregate states per episode	16×128 : the data of one episode
8	household points per state	random (ϵ, a) draws; 2×4 in the code
16,384	evaluation points per episode	$2,048 \times 8$
128 / 128	points per mini-batch / Adam steps per episode	$16,384/128$; one pass, each point once
65,000	episodes	about 8.3 million Adam steps in all, rate 10^{-5}

- ▶ All are settings of the public code: choices, not properties of the model.
- ▶ **One episode: simulate 2,048 states, pair them into 16,384 evaluation points, take 128 steps, and move on.**

Source: The continuum-of-agents code (run settings: batch size, episodes, episode length, economies, points per state, learning rate).

Deep Equilibrium Nets as an Algorithm

Deep Equilibrium Nets on Krusell–Smith

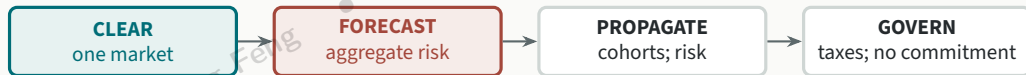
- 1 16 tracks, identical at the start: a shock and two 100-point histograms
 - 2 for each of 65,000 episodes (a setting of the code):
 - 3 draw 128 shocks; each period one lottery step per z' , store all, move to the drawn one
 - 4 store the $16 \times 128 = 2,048$ states with successors; pair each with 8 random (ε, a)
 - 5 one pass of 128 batches: Euler and KKT residuals, one Adam step each
 - 6 each track continues from its own last state in the next episode
- ▶ Intuition: if the first-order conditions hold at every state the economy visits, the policy is an equilibrium; the histogram makes the state exact.
 - ▶ **Two levers against DeepHAM: an exact state instead of a sampled one, and a one-period condition instead of a long rollout.**

Source: The paper; the continuum-of-agents code.

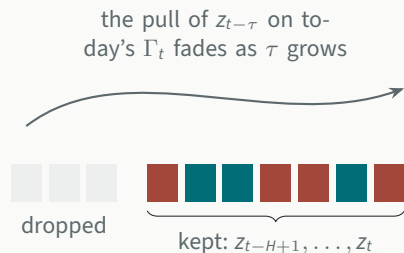
Session 4.2

Method 3: The Sequence Space

Forget the distribution: the recent shocks already determine it



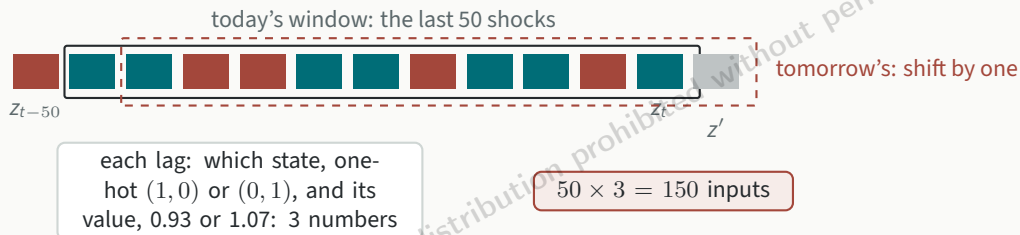
The Distribution Is a Function of Past Shocks



- ▶ Given the start and the policy, today's distribution is a function of all past shocks: $V(a, \varepsilon; \Gamma_t, z_t) = V(a, \varepsilon; z_0, \dots, z_t)$.
- ▶ **Memory loss**: old shocks fade, so keep the last H : $V \approx V(a, \varepsilon; z_{t-H+1}, \dots, z_t)$.
- ▶ The input is *exogenous*: its distribution is set by the shock process, whatever the policy does.
- ▶ **Why it works: history $\rightarrow \Gamma$ is the law of motion iterated under one fixed policy. The tutorial keeps $H = 50$.**

Source: Method 3 follows Azinovic-Yang & Žemlička, “Deep Learning in the Sequence Space”, working paper, arXiv 2509.13623 (2025), with their slides (2026) and tutorial notebook; that notebook in PyTorch: the Session 4 lab, block E.

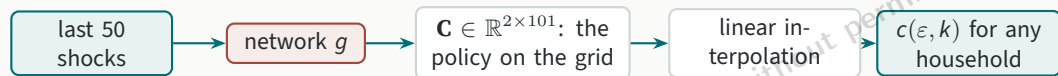
Encoding the Input: a Window of 50 Shocks



- ▶ Stored as all 50 one-hot pairs, then all 50 values: 150 numbers that stand in for the aggregate state, the distribution included. For tomorrow, drop the oldest shock and append z' .
- ▶ **Unlike DeepHAM: no household state and no moment enters; the input is exogenous, its distribution fixed by the shock process.**

Source: The tutorial notebook (history encoding and rolling windows).

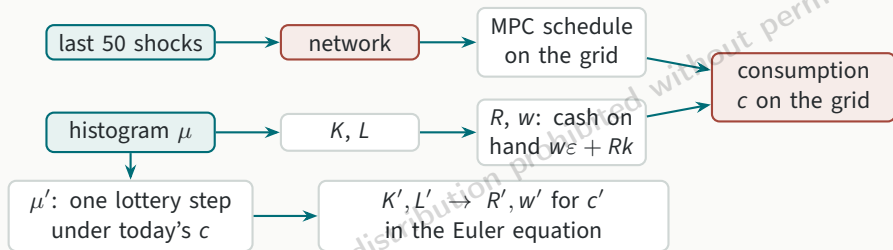
Operator Learning: History In, a Whole Policy Out



- ▶ **Operator learning** (Zhong, 2023; the paper extends it): write the policy as $f(X, x) = g(X)(x)$. The network g maps the aggregate input, here the history, to a whole *function* of the individual state $x = (\epsilon, k)$.
- ▶ The function is stored as its values on a fixed grid, and read between grid points by linear interpolation. One call gives every household's policy.
- ▶ Shape is imposed at the grid points: the output is a starting share $\alpha(\epsilon) < 1$ and falling MPCs, so c is increasing, concave and leaves $k' \geq 0$; linear interpolation keeps that shape between points.
- ▶ **Unlike DeepHAM, whose network is evaluated household by household: here the network's output is the policy function itself.**

Source: Zhong, Y. (2023), "Operator Learning in Macroeconomics", working paper; the tutorial notebook.

Where the Distribution Still Enters: Through Prices



- ▶ The network reads only shocks and gives the *shape* of consumption; the *level* needs today's prices, from μ . Tomorrow's prices need μ' , one lottery step on.
- ▶ So every state is a pair (last 50 shocks, μ), with μ simulated along those same shocks. Even a one-period loss needs a starting μ .
- ▶ **The network reads shocks; the distribution sets the prices it is evaluated at.**

Source: The tutorial notebook (the network's forward pass takes the history and today's prices; the loss).

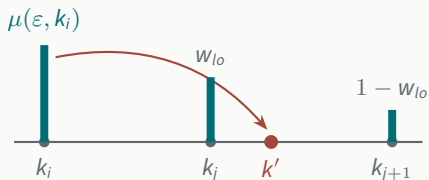
Why a Log-Spaced Grid



- ▶ The grid is where the policy function lives. Linear interpolation errs where the function bends, and consumption bends most near the borrowing limit.
- ▶ So points are dense there: equally spaced in $\log(k + 0.5)$, 0.03 apart near zero, 5 at the top. The paper suggests log-log spacing, for the same reason.
- ▶ **The histogram sits on the same grid: the lottery moves mass straight from the policy's own grid values.**

Source: The paper's section on operator learning; the tutorial notebook (capital grid).

The Same Lottery, on the Log Grid



For each ϵ and grid point k_i

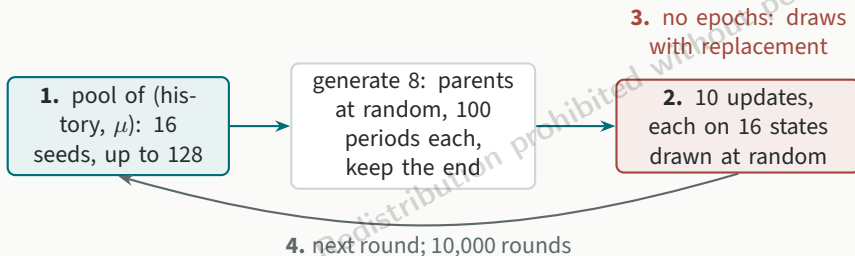
- 1 $k' = w\epsilon + Rk_i - c(\epsilon, k_i)$, clipped to $[0, 100]$
- 2 bracket: $j = \lfloor (\log(k' + 0.5) - x_0) / \Delta x \rfloor$
- 3 $w_{lo} = (k_{j+1} - k') / (k_{j+1} - k_j)$
- 4 send $w_{lo}\mu$ to k_j , the rest to k_{j+1}

$$\mu' = \Pi_{\epsilon}^{\top} \mu_{\text{after}}, \quad K' = \sum_{\epsilon, k} k \mu'(\epsilon, k), \quad L' = \sum_{\epsilon, k} \epsilon \mu'(\epsilon, k).$$

- Identical in spirit to Method 2's step; only the grid differs. In the tutorial, Π_{ϵ} does not depend on z .
- **What differs from Deep Equilibrium Nets is who reads μ' : there the network does; here only K' and L' do.**

Source: The tutorial notebook (the distribution step); Young, *JEDC* (2010).

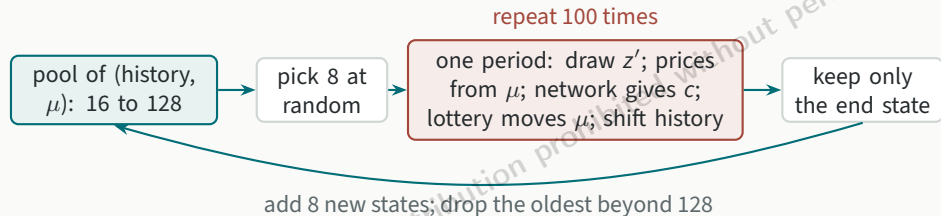
The Sequence Space's Training Loop in One Picture



- ▶ **1. Samples:** each round adds 8 states, run 100 periods from random parents.
- ▶ **2–4.** Each update draws 16 at random; no epochs; 10,000 rounds.
- ▶ **The loss looks one period ahead (both z'): it moves nothing in the pool.**

Source: The tutorial notebook (buffer, parent draws, training loop).

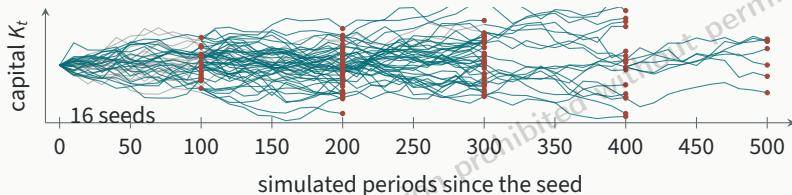
1. Samples: One Round Adds 8 States



- ▶ A state is a pair (last 50 shocks, μ); the 16 seeds are one all-low history with μ_0 . Each round, 8 states run 100 periods, network frozen; only the end state is kept.
- ▶ One round: generate 8 states, then 10 gradient steps on 16 states drawn from the pool. Pool size: 16, then 24, ..., 128 from round 14.
- ▶ **Unlike DeepHAM: 128 states in all, each a whole economy, renewed 8 at a time.**

Source: The tutorial notebook (seed, continuation, pool update).

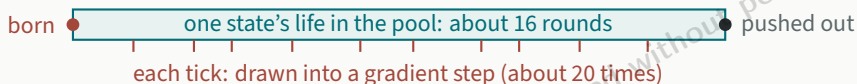
1. Samples: a Branching Pool



- ▶ Each round: 8 parents drawn at random, with replacement; each child is its parent run 100 more periods on fresh shocks. Red: the pool after 20 rounds; grey: pushed out.
- ▶ Replaying the rule for 10,000 rounds: the pool lies about 120,000 simulated periods from the seed.
- ▶ **Method 2 runs 16 fixed tracks; here tracks split and die out, and the pool is a tree's tips.**

Source: The tutorial notebook (buffer seeding, parent draws, continuation, pool update); the depth is a replay of that rule. Paths schematic.

2-4. Mini-Batches Drawn at Random; No Epochs; 10,000 Rounds



- ▶ **Mini-batch:** each update draws 16 states at random, with replacement, from the pool as it stands.
- ▶ **No epochs:** nothing is used up in order. Each round makes 10 updates; a state is drawn about 20 times in its 16 or so rounds (above).
- ▶ **Rounds:** generate 8, then 10 updates. Over 10,000 rounds: 80,000 states from 8 million simulated periods, 128 held at a time, 1.6 million state-draws.
- ▶ **Every training state was simulated under a recent version of the network.**

Source: The tutorial notebook (training loop). The paper instead moves 4,096 pairs one period per episode.

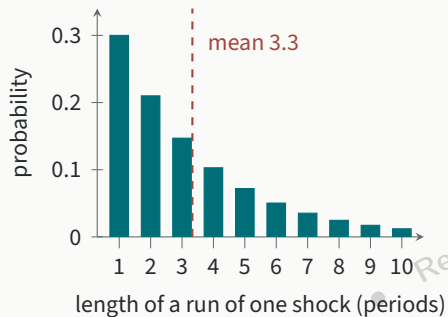
5. Inside One Update, Line by Line

One Adam step

- 1 draw 16 states (history, μ) at random from the pool built on the previous pages
 - 2 for each: R, w from μ ; one call $\mathbf{C} = g(\text{history}; R, w) \in \mathbb{R}^{2 \times 101}$; $k'_{ij} = w\varepsilon_i + Rk_j - C_{ij}$
 - 3 $\mu' =$ one lottery step of μ under c ; then K', L' , held fixed
 - 4 for each z' : $\mathbf{C}' = g(\text{shifted history}; R', w')$, read at each k'_{ij} by interpolation
 - 5 $q = \sum_{z', \varepsilon'} \pi \beta R' / c'$; then the Euler gap $e = 1/(qc) - 1$, $s = k' / c$, $\text{FB}(e, s)$
 - 6 loss = mean of FB^2 over 16×202 ; one Adam step (rate 5×10^{-5} , clipped at 1)
- ▶ One network, one call per state: its output is the whole consumption function on the grid, $c(\varepsilon_i, k_j) = C_{ij}$; not a different network per node.
 - ▶ The gradient reaches the network through $\mathbf{C}$ and $\mathbf{C}'$ only; prices, μ and μ' are held fixed.
 - ▶ **Unlike DeepHAM: one period, every grid node of every state, no rollout and no critic.**

Source: The tutorial notebook (the loss and the training step).

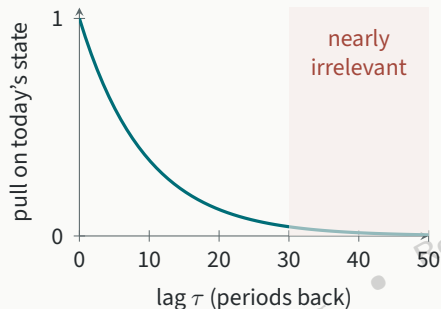
Coverage (1): The Draws Follow the True Shock Process



- ▶ All $2^{50} \approx 10^{15}$ histories are possible, and none is ruled out.
- ▶ But each new shock is drawn from the true chain: stay with probability 0.7, switch with 0.3. Training therefore meets histories as often as the economy does.
- ▶ Runs last $1/0.3 \approx 3.3$ periods on average, so likely histories are a small, structured part of the space.
- ▶ **Unlike DeepHAM and Method 2, the history's distribution does not depend on the network; only μ does.**

Source: The tutorial notebook (shock process and training loop).

Coverage (2): Old Shocks Matter Little, and μ Comes Along



- **Memory loss:** an old shock's pull on today's state decays geometrically (in the paper's one-sector example, at rate $\max\{|\rho|, \alpha\}$). Histories that differ only far back need nearly the same answer.
- μ needs no coverage of its own: it is whatever the simulated history produced.

- **Caveats:** each μ in the pool was simulated under an older network; coverage is checked on simulated paths, not guaranteed; a pool of 128 is a teaching setting.

Source: The tutorial notebook; the paper's argument that a truncated history is approximately sufficient. The decay curve is schematic.

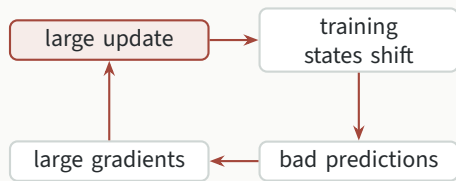
Sequence Space: the Numbers, and What Each One Is

number	what it counts	where it appears
50 / 150	shocks in the history / network inputs	the network's input: 3 numbers per shock
101	asset grid points (log-spaced)	the policy's values and the histogram share them
$2 \times 101 = 202$	grid nodes (ε, k)	one Euler residual each, per state
16	seed states	the pool at the start: all-low history, μ_0
128	pool capacity	the oldest state leaves when it is full
8 / 100	states continued per round / periods each	data generation: only the end state is kept
16	states per gradient step	drawn at random from the pool
10	gradient steps per round	after the 8 new states arrive
10,000	rounds	191 CPU-seconds in all

- **One round: generate 8 states (100 periods each), then 10 steps of 16 states $\times$ 202 residuals.**

Source: The tutorial notebook (settings, training loop, stored output).

What the Exogenous Domain Buys



state-space training can feed on itself; a history's distribution does not move with the policy

- ▶ Reported: stable training across learning rates and seeds, where the state-space version struggles.
- ▶ Tutorial: mean $|\text{FB}| 1.6 \times 10^{-4}$. Its economy: annual; ϵ independent of z .
- ▶ The cost: the policy cannot be asked about a distribution no history produced.
- ▶ **Unlike the other two: the inputs do not shift when the policy does.**

Source: Stability: their slides (2026), pp. 11, 23–24. $|\text{FB}|$: the tutorial's stored output.

Session 4.2

Three Methods Against a Law of Motion

What each method moves forward, and what a government's problem needs



How Far Forward Each Method Moves the Distribution

	what moves Γ	horizon in the loss	gradient through the move?
DeepHAM	a panel of 50, simulated	150 periods	yes, through the sampled panel
Deep Equilibrium Nets	Young's lottery on a histogram	one period	no: successors are stored data
Sequence space	Young's lottery, beside the history	one period	no: prices and K' held fixed
Feng, Han, Sargent & Zhu (2026)	the lottery matrix $\hat{\Phi}(\mathbf{x}; \theta)$, in the graph	T : 100; 30 with a government	yes, through every $\hat{\Phi}_t$

- **Methods 2 and 3 use the one-period lottery only to generate data; Feng, Han, Sargent & Zhu (2026) differentiate the whole path, DeepHAM through a sample.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the three sources' code, as on the previous pages.

Argument 1 (a): What a Markov Government Has to Compute

- ▶ Each period the government picks a tax τ ; households respond with g_τ ; successors follow the rule $\tau(\cdot)$:

$$V^G(\Gamma, z) = \max_{\tau} \left\{ W(\Gamma, z, \tau) + \beta \mathbb{E}[V^G(\Gamma', z') \mid z] \right\}, \quad \Gamma' = H(\Gamma, z, z'; g_\tau),$$

and in equilibrium the maximiser is the rule itself: $\tau(\Gamma, z)$.

- ▶ Its first-order condition:

$$\frac{\partial W}{\partial \tau} + \beta \mathbb{E} \left[\frac{\partial V^G}{\partial \Gamma'} \cdot \frac{\partial H}{\partial g} \cdot \frac{\partial g_\tau}{\partial \tau} \right] = 0.$$

- ▶ H is the law of motion, not a histogram: it maps today's distribution to tomorrow's. The second term: τ changes saving g_τ , which moves Γ' through H ; with $\beta R \approx 1$ it compounds.
- ▶ **Needed: a rule that reads Γ , households that respond to τ , and $d\Gamma'/d\tau$ through H .**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull, *RES* (2008).

Argument 1 (b): Which Method Supplies What

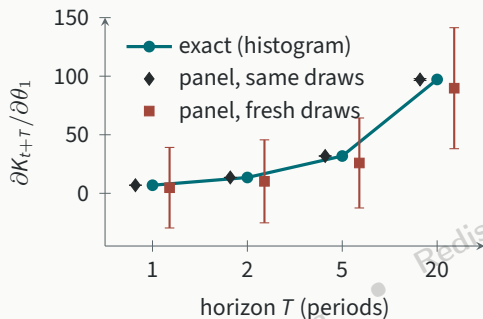
	rule reads Γ	households respond to τ	$d\Gamma'/d\tau$	successor governments
DeepHAM	through a moment of 50	tax fixed; a planner option picks c	through a sampled panel	no
Deep Eq. Nets	yes, the histogram	not in the code	no: Γ' is stored data	no
Sequence space	no: shocks only	not in the code	no: prices held fixed	no
Feng, Han, Sargent & Zhu (2026)	yes, the histogram	yes: implementability	yes: through $\hat{\Phi}$, every date	yes: a perceived rule

Feng, Han, Sargent & Zhu (2026) so far without aggregate risk: adding z is open.

- The sequence space fails at the root: history $\rightarrow \Gamma$ holds under *one fixed* policy; a deviating government breaks it. **Only a law of motion in the graph supplies the second term of the government's condition.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Han, Yang & E, arXiv 2112.14377, §2.4, §5; the three sources' code.

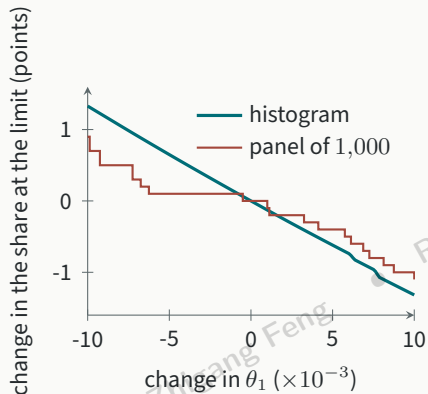
Argument 2 (a): A Derivative From a Panel, and From the Histogram



- ▶ **The test:** households save $a' = \theta_0 + \theta_1$ cash, clipped at the limit. How far does capital T periods on move per unit of θ_1 ?
- ▶ **Exact**, through the histogram: 6.90 at $T = 1$.
- ▶ **A panel** of 10,000: simulate at $\theta_1 \pm 0.001$ and difference. Fresh draws: 4.8 ± 34.4 ; same draws: 6.90 ± 0.05 . Mean $\pm$ one s.d. over 20 repetitions, as the bars.
- ▶ **For capital a panel works if its draws are fixed, as in DeepHAM's gradient.**

Source: Lab block B: $\theta_1 = 0.938$, a linear rule fitted to the solved policy; the lab's annual economy.

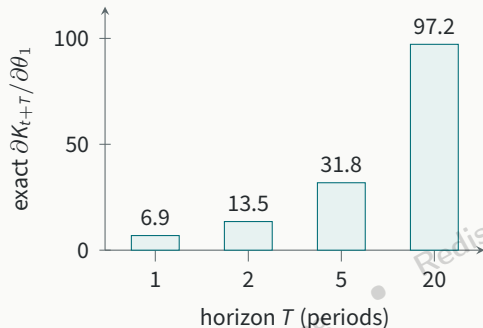
Argument 2 (b): Where a Panel Breaks: Head Counts and Clamps



- ▶ Same test, now the share at the borrowing limit after 20 periods; one household is 0.1 point.
- ▶ A step's derivative is zero wherever it exists. At $T = 2$: exact -0.13 ; panel 0.00 ± 0.00 .
- ▶ A clamped household ignores θ_1 : a panel gradient misses who is pushed onto the limit.
- ▶ **Only the lottery in the graph, moving mass $h da' / \Delta$, gives the whole derivative.**

Source: Lab block B's test swept over θ_1 , same draws at every slope; the lab's annual economy. Feng, Han, Sargent & Zhu (September 2026), preliminary.

Argument 2 (c): Why the Derivative Must Look Many Periods Ahead



- ▶ At $T = 20$ the effect is 14 times its one-period size, and still rising; five periods capture 33% of it.
- ▶ With $\beta R \approx 1$ an effect lives for about $1/(1 - \beta R)$ periods: about 420 in Deck 4.1's economy.
- ▶ A one-period condition (Methods 2 and 3) sees only the first bar.
- ▶ **DeepHAM sees 150 periods, through a sample; Feng, Han, Sargent & Zhu (2026), 100 or 30, exactly.**

Source: Lab block B, the exact histogram derivative; the lab's annual economy. Feng, Han, Sargent & Zhu (September 2026), preliminary.

Bridge: Each Tool for Its Question

Sequence space: long memory, stable training, an exogenous domain.

Deep Equilibrium Nets: the whole histogram as state; first-order conditions.

DeepHAM: interpretable moments; long rollouts.

Feng, Han, Sargent & Zhu (2026): the law of motion in the graph; a government without commitment.

- ▶ For the competitive equilibrium of this economy, all three work, and the classical rule is already accurate.
- ▶ The difference opens when someone chooses a policy that moves the distribution, and must be consistent with later choices.
- ▶ **Deck 4.3: households that age, and aggregate risk over cohorts, where the third method earns its place.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the sources of this deck.

Session 4.3

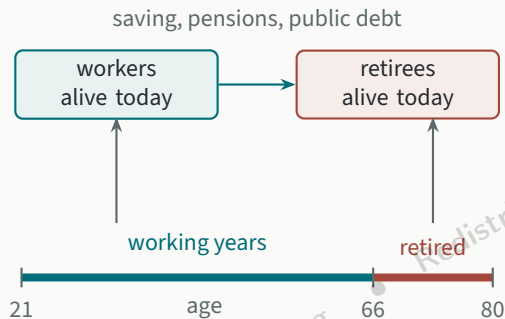
Cohorts Under Aggregate Risk

From Krusell–Smith to overlapping generations: a finite state, a network that clears a market, and why idiosyncratic risk sends us to histories



Zhigang Feng

From Krusell-Smith to Cohorts



- ▶ Deck 4.2: households live for ever; the state is $(a, \varepsilon; z, \Gamma)$ and Γ must be forecast.
- ▶ **Overlapping generations** (OLG): lives are finite, and cohorts of different ages coexist.
- ▶ Newborns hold nothing, the oldest leave, and Γ becomes a *stack*: one distribution for each age.
- ▶ **Same question as Deck 4.2, what stands for Γ , with a new structure to exploit.**

Source: Samuelson, *JPE* (1958); Diamond, *AER* (1965); Auerbach & Kotlikoff, *Dynamic Fiscal Policy* (1987).

Two Kinds of Risk, Four Economies

	no aggregate risk	aggregate risk z_t
idiosyncratic ε	Part A: the classical life cycle; Γ a stack of cohort histograms; paths after one shock	Part C: every cohort a histogram, moving with z ; the state explodes
no ε	one household per cohort, no risk: a deterministic life cycle	Part B: one household per cohort; the state is 15 numbers; a network

- ▶ Parts A and B each drop one kind of risk, and each can then carry Γ exactly.
- ▶ **Part C puts both back, and borrows Deck 4.2's third method to survive it.**

Source: Krusell & Smith, *JPE* (1998); Ríos-Rull, *RES* (1996); Azinovic-Yang & Žemlička (2025).

Session 4.3

Cohorts Without Aggregate Risk

A household that ages, a distribution indexed by age, a path of prices



Finite Lives: The Last Period Is Known

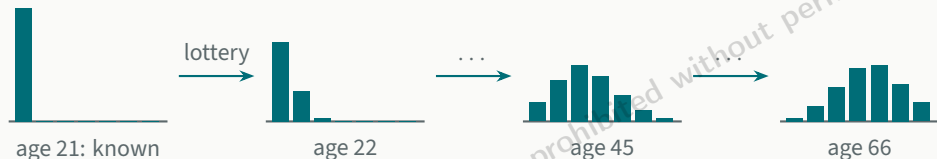


each arrow is one endogenous-grid step, using next year's policy

- ▶ The lab's life-cycle economy: 60 cohorts, ages 21 to 80, idiosyncratic ε , a limit $a' \geq 0$, a pension b .
- ▶ The policy depends on age: $a' = g_j(a, \varepsilon)$. At the last age consumption is cash on hand.
- ▶ Given c_{j+1} , Deck 4.1's Euler step delivers c_j : 59 steps in all, from age 80 back to 21. Deck 4.1's infinitely lived household instead iterates to a fixed point: 356 sweeps from a cold start ([the Session 4 lab, block A](#)).
- ▶ **Backward induction replaces iteration to convergence: the household problem has no fixed point left.**

Source: Auerbach & Kotlikoff (1987); Carroll, *Economics Letters* (2006); the Session 4 lab, block C2. Annual: $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $\rho_\varepsilon = 0.90$, $a' \geq 0$.

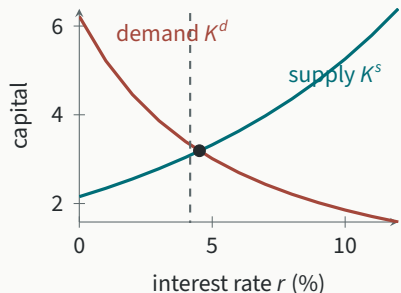
Cohorts Are the Distribution



- ▶ λ_1 is known: newborns hold nothing, with ε drawn from its invariant distribution.
- ▶ $\lambda_{j+1} = \lambda_j P_{g_j}$: Deck 4.1's *Young's Lottery*, applied once for each age.
- ▶ Given prices, no iteration to an invariant distribution: 59 pushes complete the cross-section. The price is still unknown: next page.
- ▶ **The economy's distribution is the stack of the cohorts'. Part C will need exactly this stack, moving with z .**

Source: Young, *JEDC* (2010); Ríos-Rull, *Review of Economic Studies* (1996).

The Stationary Equilibrium: One Price, Found by Bisection



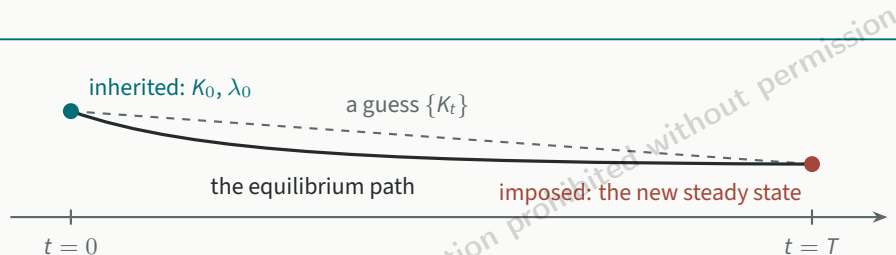
The stationary equilibrium

- 1 bracket $r \in [-7\%, 30\%]$; check the sign change
- 2 trial r : firm gives K/L , w ; tax pays pension
- 3 households backward: 59 EGM steps
- 4 cohorts forward: 59 pushes give $K^s(r)$
- 5 compare with $K^d(r)$; halve the bracket
- 6 stop after 29 halvings; residual 6×10^{-9}

- ▶ $r^* = 4.52\%$ lies above $1/\beta - 1 = 4.17\%$ (dashed): with finite lives that bound fails; the bracket must be wide.
- ▶ **One price, one fixed point; inside it, one backward and one forward pass.** Code: [the Session 4 lab, block C2.](#)

Source: The Session 4 lab, block C2: 60 cohorts, a 40% pay-as-you-go pension.

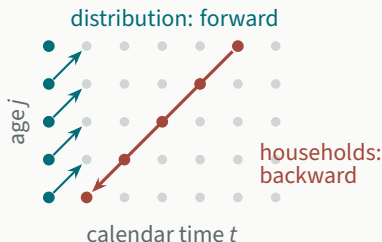
The Unknown Is a Path of Prices



- ▶ An unanticipated, permanent change at $t = 0$: an “MIT shock”. Afterwards the path is known.
- ▶ Unknown: $\{K_t\}_{t=0}^T$. Prices $\{r_t, w_t\}$ follow from the firm’s conditions.
- ▶ Boundary conditions at both ends: λ_0 is inherited, and behaviour at T is the new steady state’s.
- ▶ **Decisions look forward and distributions look back. No single pass can do both.**

Source: Auerbach & Kotlikoff (1987); Conesa & Krueger, *RED* (1999).

Update, Damp, Repeat



The transition loop

- 1 solve both steady states; choose the horizon T
- 2 guess $\{K_t\}$; prices follow from it
- 3 households backward along each cohort's diagonal
- 4 distribution forward from λ_0 : the supply K_t^s
- 5 stop when $\max_t |K_t^s - K_t|/K_t$ is small
- 6 else $K_t \leftarrow (1 - d)K_t + dK_t^s$, K_0 fixed

- Damping $d = 0.5$: a higher guess lowers r_t and the saving meant to support it, so a full step can oscillate.
- [the Session 4 lab, block C](#) runs this loop for Deck 4.1's infinitely lived economy after a 20% tax on capital income: 91 passes to 1×10^{-7} , $T = 300$. **The outer loop is a fixed point in a sequence.**

Source: Auerbach & Kotlikoff (1987); Conesa & Krueger (1999); the Session 4 lab, block C. Annual: $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $\rho_\varepsilon = 0.90$, $a' \geq 0$.

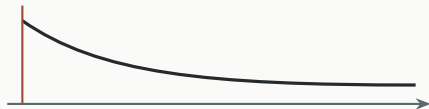
Measuring Who Gains

- ▶ **Consumption-equivalent variation** $g(a, \varepsilon)$: scale the old consumption plan by $(1 + g)$ at every date and in every state, until its value matches the new one. $g > 0$ is a gain.
- ▶ With CRRA utility V scales by $(1 + g)^{1-\sigma}$, so $g(a, \varepsilon) = [V_1(a, \varepsilon)/V_0(a, \varepsilon)]^{1/(1-\sigma)} - 1$.
- ▶ Everything turns on *which* V_1 . Along the path: the value at $t = 0$ of living through the transition. Between steady states: the value of waking up in the new one.
- ▶ Average with the distribution of the people who are asked: λ_0 , the households alive at the reform.
- ▶ **One formula, two numbers. The difference is whether the path is counted.**

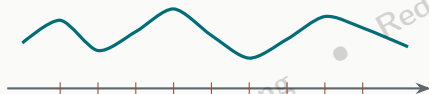
Source: Lucas, *Models of Business Cycles* (1987); Domeij & Heathcote, *International Economic Review* (2004).

One Shock Is Not a Business Cycle

one shock



Part A: a known path afterwards



a shock every period: no path to compute

- ▶ A path answers one unanticipated shock, after which everyone knows the future.
- ▶ With z_t drawn every period, households must forecast; the policy is again a function of the state (z, Γ) .
- ▶ That is Krusell and Smith's problem, now with a stack of cohort distributions.
- ▶ **Part B: drop idiosyncratic risk, and the stack becomes a short vector a network can read.**

Source: Krusell & Smith (1998); Boppart, Krusell & Mitman, *JEDC* (2018).

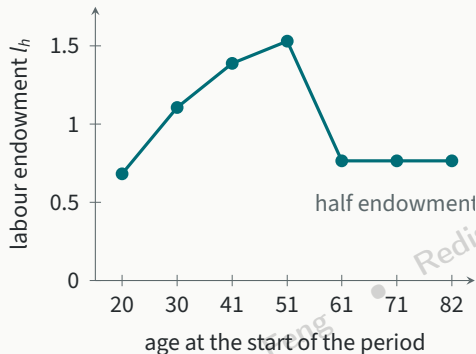
Session 4.3

Aggregate Risk Without Idiosyncratic Risk

After Azinovic-Yang & Žemlička, *Economics-Inspired Neural Networks with Stabilizing Homotopies*: without ϵ a cohort is one household, and the distribution is a vector



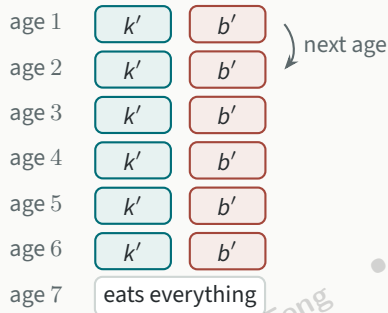
The Economy: Seven Cohorts, Two Assets



- ▶ $H = 7$ cohorts; one period is $72/7 \approx 10.3$ years, ages 20 to 82. Endowments l_h (left), summing to 7.
- ▶ CRRA $\gamma = 2$; $\beta = 0.96^{72/7}$. Two assets: capital k and a one-period bond b .
- ▶ TFP: $\log z' = \rho \log z + \sigma \epsilon$, annual $\rho = 0.85$, $\sigma = 0.03$, rescaled to the period.
- ▶ Depreciation moves with TFP: $\delta_t = \min\{\delta \frac{2}{1+z_t}, 1\}$, δ from 10% a year. **A bad draw also destroys capital.**

Source: Part B follows Azinovic-Yang & Žemlička, “Economics-Inspired Neural Networks with Stabilizing Homotopies”, arXiv 2303.14802, and its OLG notebook (*L_profile*, *Sigma_normalize*, the calibration cells). In PyTorch, scaled for a CPU: [the Session 4 lab, block F](#).

The Household: Capital, Bonds, Adjustment Costs



- ▶ Age h with holdings (k, b) chooses (k', b') :
$$c + p b' + k' + \psi_k (k' - k)^2 = l_h w + b + (1 - \delta + r) k.$$
- ▶ $k' \geq 0$; $b' \geq b_{\min} = -0.05$; the last cohort saves nothing. $\psi_k = 0.1$: changing one's own capital is costly.
- ▶ Six ages choose: 6 + 6 saving decisions.
- ▶ **No ε : everyone of an age is identical, so a cohort is one household.**

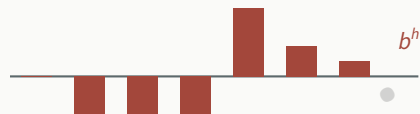
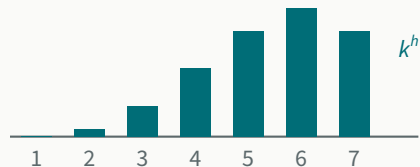
Source: Notebook markdown “Households” and the budget in E ; the constraint cell.

Firm and Markets

- ▶ Firm: $Y = z K^\alpha L^{1-\alpha}$, $\alpha = 1/3$; $r = \alpha Y/K$, $w = (1 - \alpha)Y/L$.
- ▶ Aggregates: $K = \sum_h k^h$, $L = \sum_h l_h = 7$, fixed.
- ▶ Bonds in zero net supply: $\sum_h b'^h = B = 0$. The bond price p is the one price not given by a marginal product.
- ▶ Goods clear by Walras' law.
- ▶ The notebook's text swaps the formulas for r and w ; the code, `get_firm_side`, is as written here.
- ▶ **Two assets, one of them priced by the market: a network must now return a price as well as policies.**

Source: Notebook `get_firm_side`; market-clearing markdown.

The State Is Fifteen Numbers

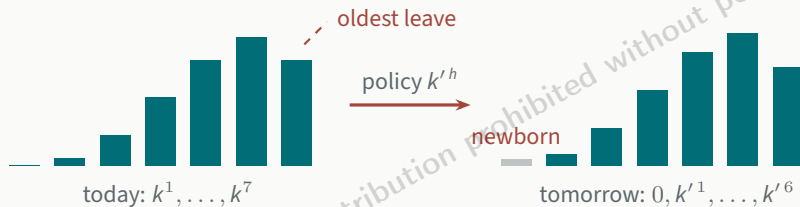


age h ; bars schematic

- ▶ $x = (z, k^1, \dots, k^7, b^1, \dots, b^7) \in \mathbb{R}^{15}$. Newborns hold nothing; the code keeps $k^1 = b^1 = 0$ anyway.
- ▶ Unknowns: 6 capital policies, 6 bond policies and the bond price, as functions of x : 13 outputs.
- ▶ Equilibrium: $12 = 6 \times 2$ KKT conditions (six ages that save, two assets each) and bond clearing, at every x .
- ▶ **Unlike Deck 4.2: Γ is exact. No moment, no histogram, no history.**

Source: Notebook “State Variable and Equilibrium Functions”; index constants for the state.

Tomorrow's Distribution Is a Shift



- ▶ Age h 's choice today is age $h + 1$'s holding tomorrow; a newborn enters with $(0, 0)$.
- ▶ The only randomness in x' is z' : one successor for each quadrature node.
- ▶ **Unlike Deck 4.2's lottery: no mass to split between grid points. The policy's output is the next state.**

Source: Notebook *ForwardStep* and the “next period wake-up assets” in E .

Equilibrium Conditions as Fischer–Burmeister Residuals

- ▶ KKT conditions for age $h < 7$, written as relative consumption errors, u'^{-1} the inverse marginal utility:

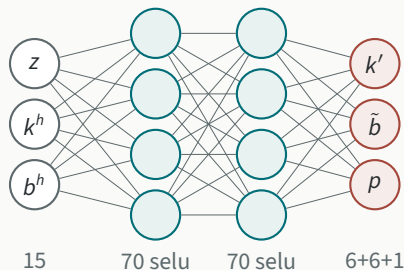
$$\varepsilon_k = \psi^{FB} \left(\frac{u'^{-1}(\beta \mathbb{E}[u'(c') (1 - \delta' + r' + 2\psi_k(k'' - k'))]) / (1 + 2\psi_k(k' - k))}{c} - 1, \frac{k'}{c} \right),$$

$$\varepsilon_b = \psi^{FB} \left(\frac{u'^{-1}(\beta \mathbb{E}[u'(c')]) / p}{c} - 1, \frac{b' - b_{\min}}{c} \right), \quad \psi^{FB}(a, b) = a + b - \sqrt{a^2 + b^2}.$$

- ▶ $\psi^{FB} = 0$ exactly when $a \geq 0, b \geq 0, ab = 0$: the Euler equation holds, or the limit binds.
- ▶ $\mathbb{E}$: 8-node Gauss–Hermite quadrature over z' . c' and k'' from the same network at x' .
- ▶ **As Deck 4.2's Deep Equilibrium Nets, one period ahead; now two assets and twelve residuals per state.**

Source: Notebook *E* (capital and bond terms), *fischer_burmeister*; Maliar, Maliar & Winant, *JME* (2021).

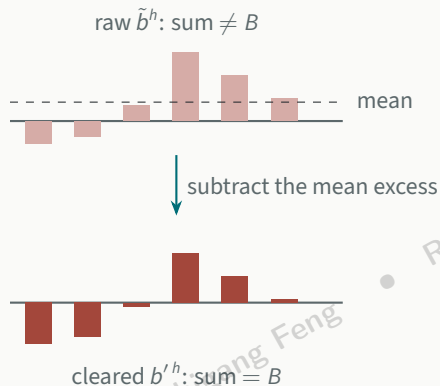
The Network: Fifteen Numbers In, Thirteen Out



- ▶ One network for all 13 functions: shared hidden layers, so ages share what they learn.
- ▶ Capital: $k' = 0.1 \text{ softplus}(o_k) \geq 0$, so no short sales by construction.
- ▶ Price: $p = \text{softplus}(o_p) > 0$.
- ▶ Bonds: the raw $\tilde{b}$ goes through a clearing layer (next page).
- ▶ **Consumption is not an output: it is backed out of the budget.**

Source: Notebook *base_net_construct*, *S_fun*; hidden width $N = 10 \times 7$.

A Layer That Clears the Bond Market



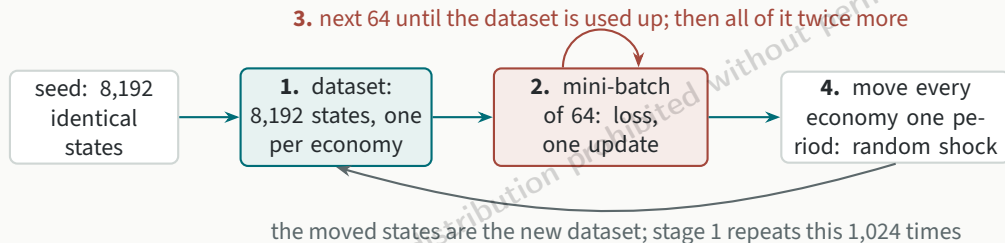
- From the six raw outputs:

$$b'^h = m \tilde{b}^h - \frac{\sum_j m \tilde{b}^j - B}{6}.$$

- Then $\sum_h b'^h = B$ holds *exactly*, at every state and every step of training.
- m is a mask the homotopy uses: $m = 0$ removes bonds altogether.
- **Unlike all three methods of Deck 4.2: a price is an output, and a market clears by construction, not by a loss term.**

Source: Notebook *S_fun*; the paper (market-clearing layers).

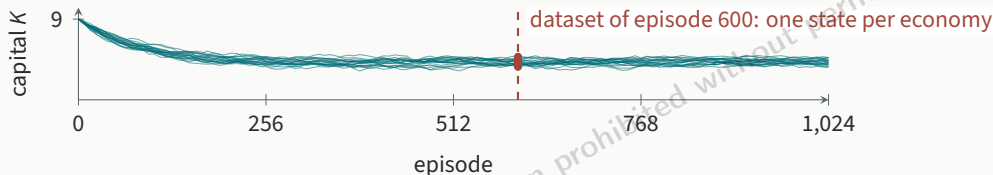
How the Network Is Trained: the Big Picture



- ▶ **1. Samples:** 8,192 economies simulated side by side; their current states are the dataset.
- ▶ **2. Mini-batch:** 64 states, the loss at them, one gradient step.
- ▶ **3. Epoch:** 128 mini-batches use the dataset once; 3 epochs per episode.
- ▶ **4. Episode:** every economy moves one period; stage 1 repeats this 1,024 times.

Source: Notebook *CloudMethod, Episode, TrainingLoop, ForwardStep*; settings *n_cloud, n_minib, epochs_epis*.

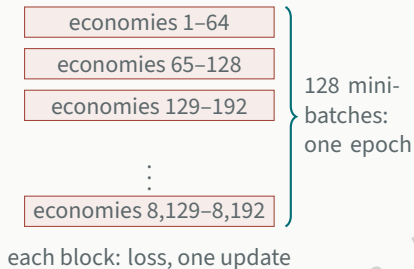
1. How the Samples Are Generated



- ▶ Each line is one economy's capital; 20 of the 8,192 are drawn. 8,192 is a setting: the number of economies simulated side by side.
- ▶ All start identical: $z = 1, k^{2..7} = 1.5$ ($K = 9$), no bonds. They drift apart only because each draws its own shocks.
- ▶ The dataset of an episode is the vertical slice: 8,192 states, one per economy, each $x = (z, k^{1..7}, b^{1..7})$.
- ▶ **Not a stored panel: only the current slice is kept, and it is replaced each episode.**

Source: Notebook settings ($n_{cloud} = 2^{13}$), the seed (*representative_input*). Paths schematic.

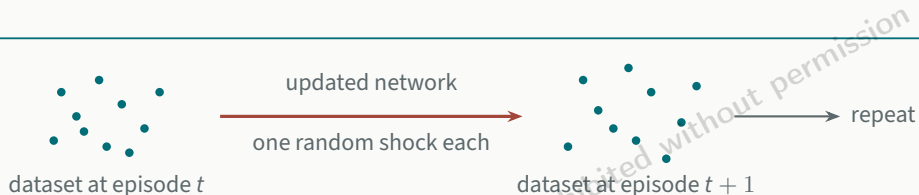
2-3. Mini-Batches, and an Epoch



- ▶ A mini-batch: 64 states taken *in order*, economies 1-64, then 65-128, and so on (64 is a setting).
- ▶ Each: the loss at its 64 states, one gradient step (Adam, 10^{-5}).
- ▶ $8,192/64 = 128$ mini-batches use the dataset up: one epoch.
- ▶ 3 epochs on the same dataset, the same 128 mini-batches each time: $3 \times 128 = 384$ updates; each state enters the loss 3 times.

Source: Notebook *TrainingLoop* ($n_{\text{minib}} = 2^6$, $\text{epochs_epis} = 3$), *Update*.

4. Episodes: the Dataset Moves One Period



- ▶ After the 3 epochs, each economy moves one period: its policy from the updated network, its own shock drawn at 2σ (the loss uses σ , so the data reach past the economy's usual range).
- ▶ The moved states replace the old ones and are the next episode's dataset.
- ▶ Stage 1, the capital-only economy, runs 1,024 episodes; the later homotopy stages continue from the same 8,192 economies.
- ▶ **This is the only simulation step. The look-ahead inside the loss (next page) computes an expectation and moves nothing.**

Source: Notebook *Episode*, *ForwardStep*, *CloudMethod*; *shock_mult* = 2 (Bretscher et al., 2022).

5. Inside One Update: One Mini-Batch, Line by Line

One update on a mini-batch of 64 states

- 1 take the next 64 states x of the dataset, in order
 - 2 for each: $K = \sum_h k^h$; r, w from z, K ; one call gives $k', \tilde{b} \rightarrow b'$ (clearing layer), p ; c from the budget
 - 3 build 8 successors $x'_j = (\rho \log z + \sqrt{2}\sigma\eta_j, 0, k'^{1..6}, 0, b'^{1..6})$, one per quadrature node η_j for z'
 - 4 call the network at each x'_j : k'', b'', p' ; then c'_j from the budget at $t + 1$
 - 5 $\mathbb{E}[\cdot] = \sum_j \omega_j(\cdot)$; the residuals $\varepsilon_k, \varepsilon_b$ for ages 1–6
 - 6 mean loss over the 64; gradient through all 9 calls, k', b' inside x'_j too; Adam, 10^{-5}
- Steps 3–5 are the look-ahead: 8 fixed quadrature nodes for z' , used only to form the expectation. They do not move the dataset. Per state, $1 + 8 = 9$ network calls; per update, $9 \times 64 = 576$.
 - **Unlike Deck 4.2's Method 2, whose successors are stored data: here they are rebuilt inside the loss, so the gradient sees the move to x' .**

Source: Notebook *E, vE, Loss, GradAndLoss, Update*.

The Loss on One Mini-Batch

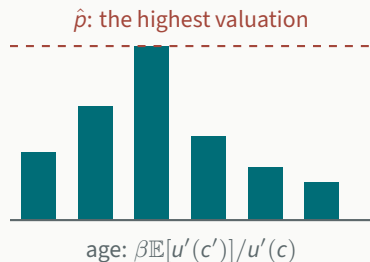
- ▶ A mini-batch B is 64 states x (the mini-batch size, 2^6 , a setting); residuals are averaged over states and ages:

$$\mathcal{L}(\theta; B) = \frac{1}{2} \overline{\varepsilon_k^2} + \frac{1}{2} w_b \overline{\varepsilon_b^2} + w_{\text{pre}} \overline{\varepsilon_{\text{pre}}^2} + 100 \overline{\varepsilon_c^2}.$$

- ▶ $\varepsilon_c = \max(c, 10^{-4}) - c$: a network that implies negative consumption is pushed back hard.
- ▶ $\varepsilon_{\text{pre}} = p - \text{sg}(\hat{p})$: a target price, with its gradient stopped.
- ▶ w_b and w_{pre} are switched by the homotopy. One Adam step per mini-batch.
- ▶ **Unlike Deck 4.2's methods, the loss weights change during training: the loss is a path of problems.**

Source: Notebook *Loss, GradAndLoss, E (eps_cnn, eps_bpr)*.

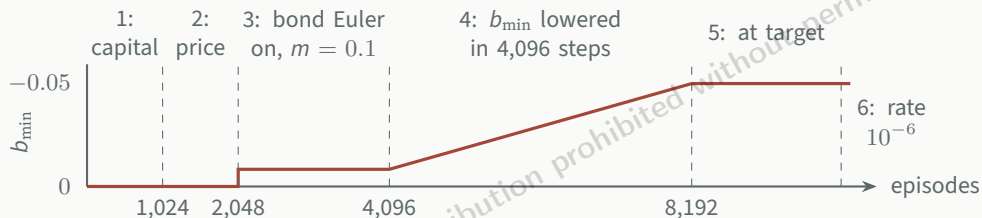
Why a Homotopy: a Bond Nobody Can Trade Has No Price



- ▶ At $b_{\min} = 0$ and $B = 0$ no one can hold a bond, so any high enough price clears the market: p is not pinned down.
- ▶ Its natural limit: the **zero-liquidity price** $\hat{p} = \max_h \beta \mathbb{E}[u'(c'^{h+1})]/u'(c^h)$, what the most eager saver would pay for the first bond.
- ▶ Then relax $b_{\min}$ a little at a time; each problem starts from the last one's solution.
- ▶ **Solving the full model from a random start is where state-space training tends to fail.**

Source: Notebook “Pretrain bond price”, $p_{b_implied}$; the paper.

The Homotopy in Six Stages



- ▶ 1: $w_b = w_{\text{pre}} = m = 0$, 1,024 episodes. 2: $w_{\text{pre}} = 1$, learn $\hat{p}$, 1,024. 3: $w_b = 1$, $w_{\text{pre}} = 0$, $m = 0.1$, $b_{\min} = -0.05/6$, 2,048.
- ▶ 4: one episode per step down to -0.05 . 5: 2,024 there. 6: 124 at a tenth of the rate. Each inherits cloud, weights, optimiser.
- ▶ **Each stage starts from the last one's network and cloud: the same 8,192 tracks run through all six, 10,340 periods.**

Source: Notebook, "Initialize Capital-Only Model" to fine-tuning; a comment says mask 0.01, the code sets 0.1.

Ten Numbers, and What Each One Is

number	what it counts	why
7	cohorts, each 72/7 years	a life cycle short enough to write out
15 / 13	network inputs / outputs	the exact state; 12 policies and a price
70	units in each of 2 hidden layers	$10 \times$ the number of cohorts
8,192	states in the cloud	wide batches instead of long simulations
64 / 384	mini-batch / Adam steps per episode	$3 \times 8,192/64$
3	passes over each cloud	then it moves one period
8	Gauss-Hermite nodes for z'	successors per state, inside the loss
2	multiple of σ when simulating	spread the training states
4,096	steps lowering $b_{\min}$ to -0.05	one episode each
$10^{-5} / 10^{-6}$	Adam rate / in the last 124 episodes	small and steady

Source: Notebook “Algorithm Choices” and the homotopy cells.

Deep Equilibrium Nets on the OLG, as an Algorithm

The homotopy notebook

- 1 seed a cloud of 8,192 identical states; a random network
 - 2 for each stage (capital; price; bonds on; lower $b_{\min}$; target; fine-tune):
 - 3 set the homotopy: $b_{\min}$, mask m , weights w_b, w_{pre}
 - 4 for each episode of the stage:
 - 5 384 Adam steps: 64 states, 9 network calls each, FB residuals
 - 6 move the whole cloud one period with fresh shocks at 2σ
- ▶ About 10,300 episodes, or four million Adam steps; the notebook times each stage in minutes on one GPU.
 - ▶ **The same loop as Deck 4.2's Deep Equilibrium Nets, with three changes: an exact state, a clearing layer, and a path of problems.**

Source: Notebook *CloudMethod*, *Episode*, and the homotopy loop.

Four Methods, One Table

	DeepHAM	Deep Eq. Nets, KS	Sequence space	Deep Eq. Nets, OLG (here)
what stands for Γ	learned moments of 50 households	two 100-point histograms	the last 50 shocks	Γ itself: 14 numbers
what is minimised	minus lifetime utility; a value regression	Euler and KKT residuals	an FB Euler residual	FB residuals; bond clearing built in
where states come from	panels, refreshed	16 economies, carried forward	a replay buffer	a cloud of 8,192, moved one period
what the gradient passes through	150 periods of the panel	one period, at stored states	one period; prices fixed	one period, through the computed x'

► **Row 1 is why the OLG is easy here: with no ε , nothing has to stand for Γ .**

Source: Deck 4.2, *Four Questions Every Deep Method Answers*; the homotopy notebook.

What Changed From Deck 4.2

- ▶ **An exact state.** Cohorts without ε are points, so the distribution is a vector, and its law of motion is a shift.
- ▶ **A price as an output.** Two assets need a bond price; the network returns it, and a layer clears the market exactly.
- ▶ **A path of problems.** Capital first, then a price, then bonds that trade a little more each step.
- ▶ **Wide and shallow simulation.** 8,192 economies moved one period at a time, at twice the shock's s.d.
- ▶ **Successors differentiated.** The loss builds x' from the policy; the gradient sees the move.
- ▶ **All five rest on the first. Take it away and the rest has nothing to stand on.**

Source: The homotopy notebook; Deck 4.2.

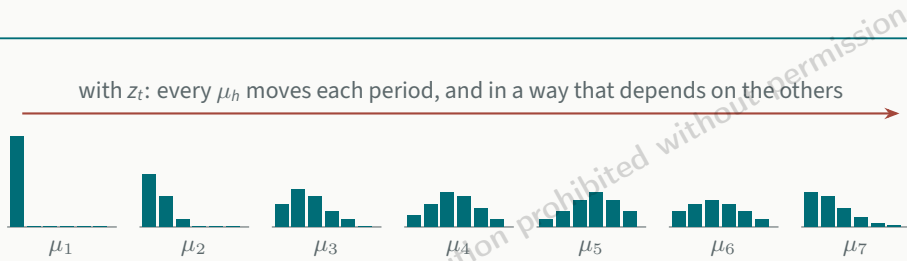
Session 4.3

Put Idiosyncratic Risk Back

After Azinovic-Yang & Žemlička, *Deep Learning in the Sequence Space*: every cohort becomes a distribution, and the state explodes



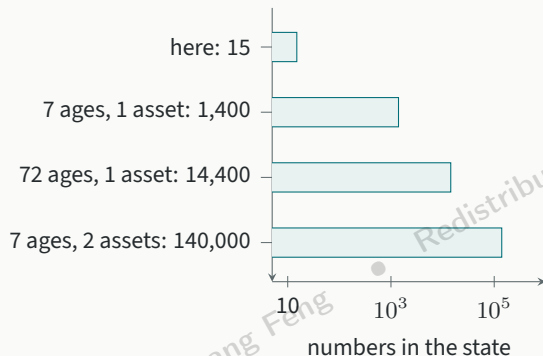
Each Cohort Becomes a Histogram



- ▶ With ε , two households of the same age differ. A cohort is a distribution $\mu_h(\varepsilon, k)$ again, as in Part A.
- ▶ The aggregate state: $(z, \mu_1, \dots, \mu_H)$. Part A's stack, but no longer stationary: it moves with z .
- ▶ **Part A had the stack without z ; Part B had z without the stack. Now both.**

Source: Part C follows Azinovic-Yang & Žemlička, “Deep Learning in the Sequence Space”, working paper, arXiv 2509.13623 (2025), §6, with their slides (2026); here p. 113.

Count the Inputs



Source: Arithmetic. Cohort count: their slides, p. 13.

- ▶ Two income states, 100 asset points per cohort: $7 \times 2 \times 100 = 1,400$.
- ▶ With the published OLG's 72 cohorts: $72 \times 2 \times 100 = 14,400$.
- ▶ Two assets on a 100×100 grid: $7 \times 2 \times 10^4 = 140,000$.
- ▶ **A state-space network reads all of it, and its training states move whenever the policy does.**

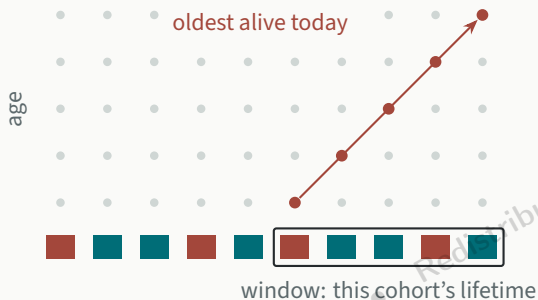
Why Deck 4.2's First Two Answers Strain

	Krusell–Smith, one distribution	OLG, H distributions
classical rule	forecast K' from K	forecast every cohort's moments, or prices from many of them
DeepHAM	one panel of 50, one learned moment	a panel for each age; moments per age
Deep Eq. Nets	2×100 masses as input	$H \times 2 \times 100$ masses, drifting with the policy
sequence space	the last 50 shocks	the same window of shocks

- ▶ The first three grow with the number of ages; the last does not, because its input never contained Γ .
- ▶ Deck 4.2, *What the Exogenous Domain Buys*: large inputs that shift with the policy feed the unstable loop.
- ▶ **Only the history keeps its size when cohorts are added.**

Source: Deck 4.2; their slides, pp. 11, 23–24.

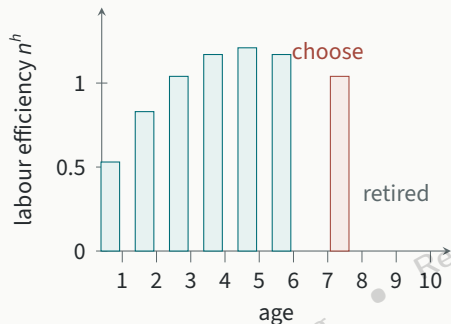
The Way Out: the History Already Holds Every Cohort



- ▶ Everyone alive was born within the last H periods: μ_h is built by the shocks of its lifetime, from its starting wealth.
- ▶ Older shocks act only through prices and inheritances (bequests, in the model ahead): Deck 4.2's **memory loss**.
- ▶ So the stack is, approximately, a function of recent shocks: a window at least a lifetime long.
- ▶ **Published sweep, 1 to 256 lags: accuracy improves with the window.**

Source: Intuition ours; evidence: their slides, pp. 7–8, 20–21, 91 (the paper's 72-cohort OLG).

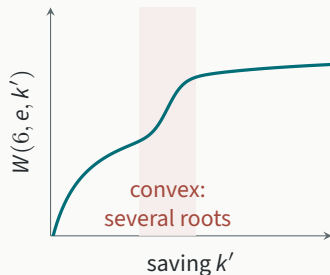
The Economy: Ten Cohorts, Income Risk, a Retirement Choice



- ▶ 10 cohorts of six years: ages 1–6 work, age 7 chooses to work or retire (logit, taste-shock scale 0.08), ages 8–10 are retired. Bequests become newborns' assets.
- ▶ Income $e \in \{0.5, 1, 1.5\}$, Markov; one asset, capital, $k \geq 0$.
- ▶ Aggregate risk: three regimes z ; the third is a disaster, depreciation 0.397 instead of 0.265.
- ▶ State (z, μ) , $\mu(h, e, k)$ on $10 \times 3 \times 300$: 9,000 masses. Input: the last 100 regimes.
- ▶ **The network never reads μ .**

Source: The paper, §6.1–6.2; their slides, pp. 105–113. The paper numbers ages 0–9; its implementation text says 200 asset points, its parameter table 300.

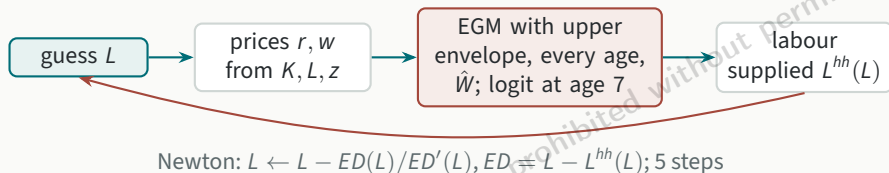
Why the Network Outputs Continuation Values



- ▶ Age 6 looks ahead to retirement, so its value of saving is convex in places (left, schematic): the Euler equation has several roots, and its residual can pick the wrong one.
- ▶ So the network outputs **continuation values** $\hat{W}(h, e, k') \approx \mathbb{E}[V(h+1, e', k')]$ on the grid, ages 1–9; EGM with an **upper envelope** keeps the best root.
- ▶ Increasing by construction: a first value plus softplus increments. One call gives all $9 \times 3 \times 300$: operator learning.
- ▶ **Unlike Deck 4.2's tutorial: no consumption output, no FB residual.**

Source: The paper, §6.3; their slides, pp. 110, 114–120; Fella (2014); Druedahl & Jørgensen (2017).

Within the Period: From $\hat{W}$ to Policies, and a Labour Market That Must Clear



- ▶ K is fixed by μ ; L is not: whether age 7 works depends on the wage, and the wage on L .
- ▶ Five Newton steps, ED' by automatic differentiation, from $L^{\text{work}} + \hat{\lambda} L^{7, \text{max}}$: $\hat{\lambda}$ is the network's guess of the share of age 7's labour supplied.
- ▶ Result: prices, every age's saving rule $g^h(e, k)$, the work probabilities $p^{\text{work}}(e, k)$, and the realised share λ .
- ▶ **Unlike Part B, where a layer cleared the bond market exactly: here a price is found by iteration inside every evaluation.**

Source: The paper, §6.3 (Step 0 of the forward simulation).

Moving the Distribution One Period

- ▶ Young's lottery with income transitions, for a saving rule g and masses $m(i, j)$ at (e_i, k_j) :

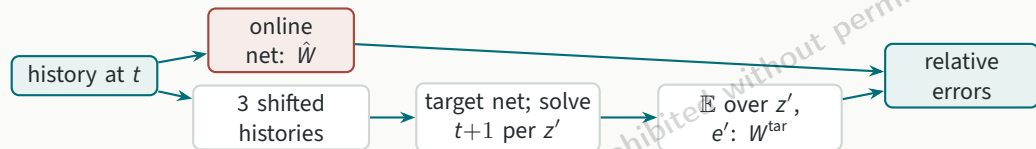
$$[\mathcal{T}(g, m)](i', j') = \sum_{i, j} m(i, j) \Pi_{ii'}^e (\omega^- \mathbf{1}\{j' = j^-\} + \omega^+ \mathbf{1}\{j' = j^+\}),$$

$j^\pm$ the grid points around $g(e_i, k_j)$, $\omega^\pm$ the lottery weights of Deck 4.2.

- ▶ Aging: $\mu'_{h+1} = \mathcal{T}(g^h, \mu_h)$. Age 7 splits: $\mathcal{T}(g^{\text{work}}, p^{\text{work}} \odot \mu_7) + \mathcal{T}(g^{\text{ret}}, (1 - p^{\text{work}}) \odot \mu_7)$.
- ▶ Newborns: $\mu'_1 = \mathcal{T}(g^{10}, \mu_{10})$, the bequests.
- ▶ μ' does not depend on z' : three candidate successors (z', μ') , each with its history shifted by z' .
- ▶ **The same lottery as Deck 4.2's Methods 2 and 3, applied age by age.**

Source: The paper, §6.3 (Steps 1–3 of the forward simulation); Young, *JEDC* (2010).

The Loss: Targets From a Slower Copy of the Network



- ▶ Target: $W^{\text{tar}}(h, e, k') = \sum_{z'} \Pi_{zz'}^z \sum_{e'} \Pi_{ee'}^e V_{t+1}(h+1, e', k' \mid z'; \bar{\rho})$, $t+1$ solved with the target network $\bar{\rho}$.
- ▶ Loss: mean squared relative errors, $(\text{target} - \text{prediction}) / (\epsilon_{\text{loss}} + |\text{target}|)$, on W at the first grid point, on W 's increments, and on λ ; ϵ_{loss} from 1 to 0.1 to 0.01.
- ▶ Targets fixed; gradient through the online network only; then $\bar{\rho} \leftarrow \tau \rho + (1 - \tau) \bar{\rho}$, $\tau = 10^{-4}$.
- ▶ **Unlike the tutorial's residual: a supervised target from a slower copy, as Deck 4.1's critic.**

Source: The paper, §6.3 (loss function) and §6.4 (Stage 3).

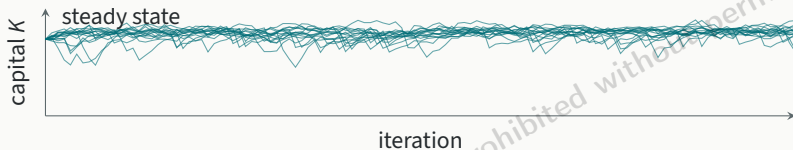
One Iteration, Line by Line

One Adam step on 1,024 pairs (z, μ , history)

- 1 online network at each history: $\hat{W}$ and $\hat{\lambda}$
 - 2 target network at the 3 shifted histories
 - 3 clear labour at t (5 Newton steps, each an EGM solve): prices, policies, λ
 - 4 move μ one period: 3 candidate successors
 - 5 solve $t+1$ for each z' with the target network; take the expectation: W^{tar}
 - 6 loss; one Adam step at 10^{-6} ; update the target network
 - 7 draw z' from Π^z ; that successor is the pair's next state
- ▶ Three stages: (1) a deterministic steady state by EGM, which seeds the pairs; (2) aggregate risk on, continuation values held at the steady state, to spread the pairs; (3) the loop above.
 - ▶ Network: three hidden layers of 1,024 (GELU); history length 100.

Source: The paper, §6.4 (outline, Stages 1–3, hyperparameters).

Where the 1,024 Pairs Come From, and What They Cover



- ▶ Start: the steady-state distribution. Each iteration moves every pair one period: z' from Π^z , μ' by the lottery, the history shifted. Dips: disaster regimes.
- ▶ As in Part B, many tracks, one period per step; unlike Deck 4.2's Method 3, no pool and no branching.
- ▶ Coverage: histories are drawn from Π^z , so the 3^{100} windows are met at their true frequencies; a life is 10 periods, the window 100. μ comes along with its history.
- ▶ **Coverage is checked on simulated paths, not guaranteed.**

Source: The paper, §6.3 (Step 3) and §6.4 (step 7). Paths schematic.

Bridge: Each Tool for Its Question

Paths: one unanticipated shock, any distribution.

The state as input: when Γ is a short vector, as here.

The history as input: when Γ is many distributions.

Next: back to Deck 4.1's histogram, and a government that taxes on it.

- ▶ Age turns Γ into a stack. Without ε the stack is a vector, and Deep Equilibrium Nets solve it exactly.
- ▶ With ε the stack is many histograms: the history is the input, continuation values the output, EGM the solver.
- ▶ **Deck 4.4: a government that cannot commit chooses a tax rule on the whole distribution.**

Source: The sources of this deck.

Session 4.4

Optimal Taxation Without Commitment

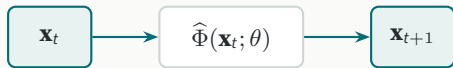
Put a government into Deck 4.1's economy: a tax rule on the whole distribution, a Markov-perfect fixed point in rules, and the algorithm that finds it



Zhigang Feng

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Where Decks 4.1 and 4.2 Stopped



Deck 4.1, a household: $\mathbf{x}_{t+1}$ taken as given



this deck, a government: it sees $d\mathbf{x}_{t+1}$

- ▶ Deck 4.1: the histogram $\mathbf{x}$ and its matrix $\hat{\Phi}$, rebuilt at every date inside the graph. A household is atomistic: it takes $\mathbf{x}_{t+1}$ as given.
- ▶ Deck 4.2, *Argument 1*: a government that chooses its tax every period needs $d\Gamma'/d\tau$, and only a law of motion in the graph supplies it. Here τ is a function of $\mathbf{x}$, so households respond to it through $\mathbf{x}$.
- ▶ This deck puts that government into Deck 4.1's economy.
- ▶ **One object is the state, the law of motion and the gradient's path.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Decks 4.1 and 4.2.

The Question: a Tax Nobody Can Promise



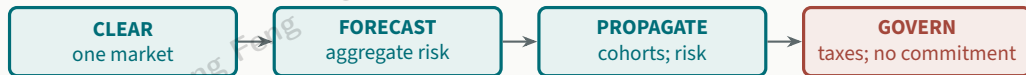
- ▶ Under commitment the long-run capital tax is zero in the classic models: promise not to tax capital, and people save.
- ▶ Capital in place is an inelastic base. A government that chooses afresh each period is tempted to tax it; savers know it.
- ▶ Without commitment the problem is a game between today's government and its successors.
- ▶ The answer is a rule on the whole distribution, $\tau(\Gamma)$: not a number, not a function of a few moments.
- ▶ **Sought: the tax a government chooses when every successor will choose by the same rule.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Chamley (1986); Judd (1985); Klein, Krusell & Ríos-Rull, *RES* (2008).

Session 4.4

The Economy and Its Equilibrium

Deck 4.1's households, a government that taxes and cannot commit, and a fixed point in rules



Households

- ▶ As in Deck 4.1: ε redrawn every period (i.i.d.), $Q = 5$ Gauss–Hermite nodes (ε_q, p_q) , r net of depreciation. A household maximises

$$\sum_{t=0}^{\infty} \beta^t U(c_t, l_t, G_t), \quad U(c, l, G) = u\left(c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}\right) + \gamma_g \log G, \quad u(y) = \frac{y^{1-\sigma}}{1-\sigma},$$

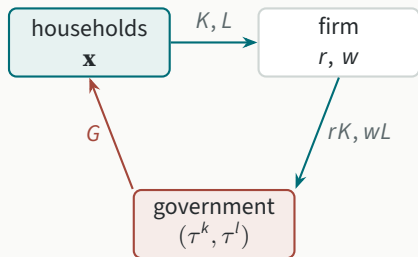
subject to

$$c_t + a_{t+1} = [1 + (1 - \tau_t^k) r_t] a_t + (1 - \tau_t^l) w_t \varepsilon_t l_t, \quad a_{t+1} \geq 0.$$

- ▶ τ^k taxes capital income, τ^l labour income; G is a public good, valued with weight γ_g .
- ▶ G enters additively, so it changes no household's choice: only the taxes do.
- ▶ **Hours still carry no wealth effect: the Frisch elasticity is $1/\theta_l = 3.125$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\beta = 0.95$, $\sigma = 2$, $\theta_l = 0.32$, $\psi_l = 1.5$, $\sigma_\varepsilon = 0.2$.

Firms, a Balanced Budget, Market Clearing

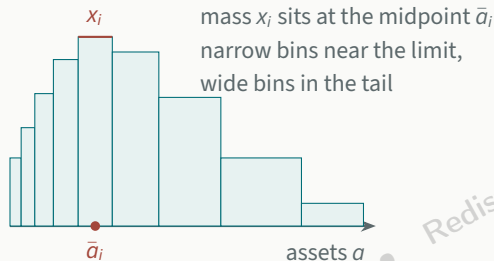


no bonds: G is whatever the taxes raise this period

- ▶ Firm: $Y = K^\alpha L^{1-\alpha}$, $r = \alpha(K/L)^{\alpha-1} - \delta$, $w = (1 - \alpha)(K/L)^\alpha$.
- ▶ Government: picks (τ^k, τ^l) and balances its budget every period, $G = \tau^k rK + \tau^l wL$.
- ▶ Markets: $K = \sum_i x_i \bar{a}_i$, $L = \mathbb{E}[\varepsilon l(\varepsilon)]$, exactly; the histogram moves by $\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}_t$.
- ▶ No bonds and no commitment technology: today's government cannot bind its successors.
- ▶ **G is a residual, not a choice: two instruments, one budget, every period.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\alpha = 0.34$, $\delta = 0.06$.

The Distribution, Approximated by a Histogram



- ▶ $N = 140$ bins on $[0, 16]$, midpoints $\bar{a}_1 < \dots < \bar{a}_N$; the state is the row vector $\mathbf{x} = (x_1, \dots, x_N)$, $x_i \geq 0$, $\sum_i x_i = 1$.
- ▶ Aggregates are sums: $K = \sum_i x_i \bar{a}_i$.
- ▶ ε is i.i.d.: $Q = 5$ Gauss-Hermite nodes (ε_q, p_q) integrate it, so $\mathbf{x}$ needs no ε dimension.
- ▶ **The government's state is these N numbers.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; as in Deck 4.1. The grid is fixed once, from the classical benchmark.

The Law of Motion $\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}_t$, Built Entry by Entry

At date t , for every source bin i and node q

- 1 saving at the midpoint: $a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$, kept inside $[\bar{a}_1, \bar{a}_N]$
- 2 bracket it, $\bar{a}_j \leq a'_{t,iq} \leq \bar{a}_{j+1}$ with $j = j_{t,iq}$, and weigh the upper neighbour: $\omega_{t,iq} = \frac{a'_{t,iq} - \bar{a}_j}{\bar{a}_{j+1} - \bar{a}_j}$
- 3 send $p_q(1 - \omega_{t,iq})$ of bin i 's mass to bin j and $p_q \omega_{t,iq}$ to bin $j + 1$ (Young's lottery)

$$\hat{\Phi}_{t,ij'} = \sum_{q=1}^Q p_q \left[(1 - \omega_{t,iq}) \mathbf{1}\{j' = j_{t,iq}\} + \omega_{t,iq} \mathbf{1}\{j' = j_{t,iq} + 1\} \right], \quad x_{t+1,j'} = \sum_{i=1}^N x_{t,i} \hat{\Phi}_{t,ij'}.$$

- Row i says where bin i 's households go: rows sum to one, so mass is conserved.
- Exact in the mean: $\sum_{j'} \hat{\Phi}_{t,ij'} \bar{a}_{j'} = \sum_q p_q a'_{t,iq}$, so K_{t+1} equals what households save.
- $\hat{\Phi}_t$ moves with θ and with $\mathbf{x}_t$, through a' : it is rebuilt at every date, and ω is linear in a' , so it can be differentiated.

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Young, JEDC (2010); Deck 4.1. $N \times Q$ network calls per date; no simulation, no sampling noise.

Household Conditions

- ▶ Saving: the Euler condition with the limit, equality whenever $a' > 0$,

$$u'(y_t) \geq \beta \mathbb{E}_{\varepsilon'} \left\{ [1 + (1 - \tau_{t+1}^k) r_{t+1}] u'(y_{t+1}) \right\}, \quad y = c - \psi_l \frac{l^{1+\theta_l}}{1 + \theta_l}.$$

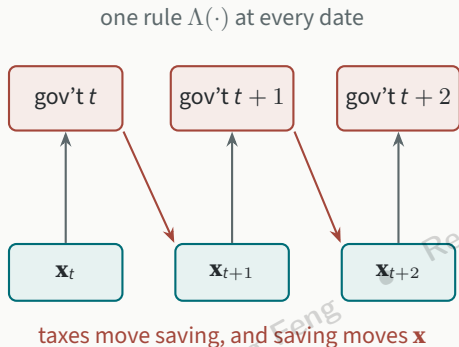
- ▶ Hours, in closed form:

$$(1 - \tau^l) w \varepsilon = \psi_l l^{\theta_l} \implies l(\varepsilon) = \left(\frac{(1 - \tau^l) w \varepsilon}{\psi_l} \right)^{1/\theta_l}.$$

- ▶ *Tomorrow's capital tax sits inside today's saving decision: expectations about successors move the economy now.*
- ▶ Hours contain no α . Given τ^l and K , labour L , both prices and both tax bases are fixed, whatever the shape of $\mathbf{x}$.
- ▶ **Under GHH the marginal utility of consumption is $u'(y)$, so $U_c = u'(y)$ in every condition below.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Greenwood, Hercowitz & Huffman, *AER* (1988).

State, Rule, Timing



- ▶ State: $(\varepsilon, a; \mathbf{x})$ for a household; $\mathbf{x}$ alone for the government.
- ▶ A **Markov rule** $\Lambda(\mathbf{x}) = (\tau^k(\mathbf{x}), \tau^l(\mathbf{x}))$, with $G(\mathbf{x})$ from the budget: no calendar time, no history.
- ▶ Every government, at every date, uses the same rule, and that rule is the unknown.
- ▶ Households take a perceived rule $\tilde{\Lambda}$ and law of motion Φ as given, and best-respond.
- ▶ **The Markov restriction, not foresight, is what makes the rule time-consistent.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull (2008).

The Two Problems

(P-1) Household. Given prices, G , the perceived rule $\tilde{\Lambda}$ and $\mathbf{x}_+ = \mathbf{x}\hat{\Phi}$,

$$V(\varepsilon, a; \mathbf{x}) = \max_{0 \leq a'} \left\{ U(c, l, G) + \beta \sum_{q'} p_{q'} V(\varepsilon_{q'}, a'; \mathbf{x}_+) \right\}.$$

(P-2) Government (Deck 4.2's V^G). Given the successors' rule $\tilde{\Lambda}$ and its value $\tilde{W}$,

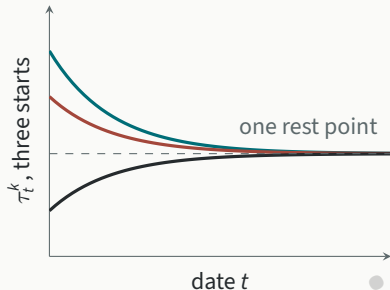
$$W(\mathbf{x}) = \max_{\tau^k, \tau^l} \left\{ \sum_{i,q} x_i p_q U(c_{iq}, l_q, G) + \beta \tilde{W}(\mathbf{x}_+) \right\},$$

subject to the budget, prices, the household's conditions at every node $(\bar{a}_i, \varepsilon_q)$, and $\mathbf{x}_+ = \mathbf{x}\hat{\Phi}$.

- ▶ Welfare is utilitarian: each node counts by its mass $x_i p_q$.
- ▶ **The asymmetry: the household takes $\mathbf{x}_+$ as given; the government is not atomistic and sees how its choice moves $\mathbf{x}_+$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull (2008).

Markov-Perfect Equilibrium



- ▶ A **Markov-perfect equilibrium**: saving and value functions, prices, a rule Λ with value W , and perceptions, with
 - (a) households solving (P-1); (b) $\Lambda(\mathbf{x})$ solving (P-2) at every $\mathbf{x}$; (c) perceived = actual: $\tilde{\Lambda} = \Lambda$, $\tilde{W} = W$, Φ from households' saving; (d) markets clearing.
- ▶ No aggregate shocks: from *any* $\mathbf{x}_0$, taxes and aggregates must settle to constants.
- ▶ **A sharp test: paths from different starts must meet.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic curves: direction and shape only.

Three Computational Challenges

Challenge	Deck 4.2's methods	Here
the distribution is a state	learned moments of a panel; a histogram; a history of shocks	Deck 4.1's histogram, and quadrature for ε
inequality constraints	Fischer–Burmeister: two inequalities become one equation	the same, at every node, priced by a penalty
strategic interaction	none of the three has a government that re-chooses	Deck 4.1's matrix, with the gradient passed through it

- ▶ Simulating Γ_+ works when nobody acts on the distribution; with a player who does, it breaks the forward and backward passes (Deck 4.2, *Argument 2*).
- ▶ **Deck 4.1 settled the first row. This deck takes the other two, then asks: is the answer right?**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar et al. (2021); Azinovic et al. (2022); Han, Yang & E (2026).

Implementability: One Equation per Household State

- ▶ At every node $(\bar{a}_i, \varepsilon_q)$ the government must respect the Euler condition with the limit, as one equation:

$$\begin{aligned} \text{FB}(\vartheta_1, \vartheta_2) &= \vartheta_1 + \vartheta_2 - \sqrt{\vartheta_1^2 + \vartheta_2^2 + \eta^2} = 0, & \bar{E}_{iq} &= \text{FB}_{iq} / u'(y_{iq}), \\ \vartheta_1 &= u'(y) - \beta \sum_{q'} p_{q'} [1 + (1 - \bar{\tau}^k(\mathbf{x}_+)) r(\mathbf{x}_+)] u'(y_{+,q'}), & \vartheta_2 &= a'. \end{aligned}$$

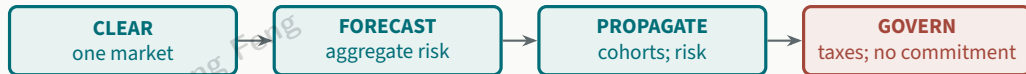
- ▶ $\text{FB} = 0$ when both are non-negative and one is zero; $\eta = 10^{-4}$. $\bar{E}$ is a relative Euler error.
- ▶ a' is the saving *after* the clamp, as the budget and $\hat{\Phi}$ use it (Deck 4.1: pre-clamp $\tilde{a}$).
- ▶ Tomorrow's tax is the *perceived* rule $\bar{\tau}^k$, at the $\mathbf{x}_+$ today's saving produces.
- ▶ **The function is held fixed; the state it is evaluated at is not. There the strategic channel lives.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar, Maliar & Winant (2021).

Session 4.4

The Ingredients and the Algorithm

Three networks, one penalised objective, a rollout through the matrix, and a frozen copy of the future



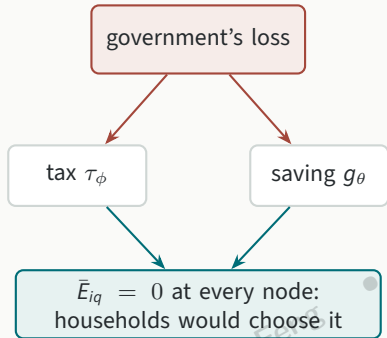
Three Trained Networks

Network	Input	Out	Head
saving g_θ	$(\varepsilon, a, \mathbf{x})$	1	Deck 4.1's change head, $\tilde{a} = a + D(2s - 1)$, $a' = \min\{\max\{\tilde{a}, 0\}, \bar{a}\}$
tax rule τ_ϕ	$\mathbf{x}$	2	a sigmoid times a cap: $\tau^k = \tau_{\max}^k \sigma(\cdot)$, $\tau^l = \tau_{\max}^l \sigma(\cdot)$
aggregate value $\bar{V}_\chi$	$\mathbf{x}$	1	linear: the government's value beyond the rollout
household critic V_ψ	$(\varepsilon, a, \mathbf{x})$	1	Deck 4.1's critic: loaded, not trained here

- ▶ Each: two hidden layers of 256 rectified-linear units. The tax rule reads the whole histogram, so it can respond to its *shape*.
- ▶ **The saving network reads $\mathbf{x}$, not the tax: in equilibrium the tax is itself a function of $\mathbf{x}$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $D = 1.3$, as in Deck 4.1.

The Government Chooses the Allocation Too

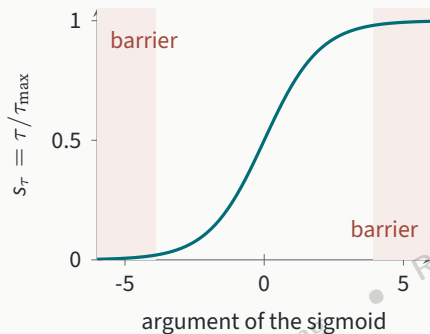


one backward pass,
two optimiser steps

- ▶ Deck 4.1 trained saving on the household's own Bellman equation. Here it learns only from the government's loss.
- ▶ The government picks the tax *and* the saving rule; $\bar{E} = 0$ makes it one household would choose.
- ▶ The **primal approach** of Ramsey taxation, with a Markov government in place of a committed one.
- ▶ **The rule is the households' only as far as the penalty drives $\bar{E}$ to zero.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Chari & Kehoe, *Handbook of Macroeconomics* (1999), ch. 26.

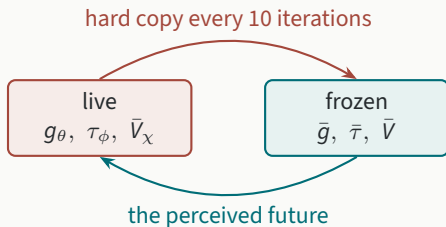
The Tax Rule's Head Decides What Can Be Learned



- ▶ $\tau^k(\mathbf{x}) = \tau_{\max}^k s_\tau$, $s_\tau = \sigma(\ell(\mathbf{x}))$, ℓ the last layer: its slope carries $\tau_{\max}^k s_\tau (1 - s_\tau)$.
- ▶ At the cap, $s_\tau \rightarrow 1$ and the gradient vanishes. A constant tax is a trivial fixed point: the run looks converged.
- ▶ Remedy one: a cap $\tau_{\max}$ wide enough for an interior optimum. A solver setting, not economics.
- ▶ Remedy two: a hinge barrier, zero inside, $B_\sigma = \sum [(0.02 - s_\tau)_+^2 + (s_\tau - 0.98)_+^2]$.
- ▶ **A floor under the gradient wherever τ lands, or the rule cannot learn to respond to $\mathbf{x}$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Perceived Equals Actual: the Frozen Copy

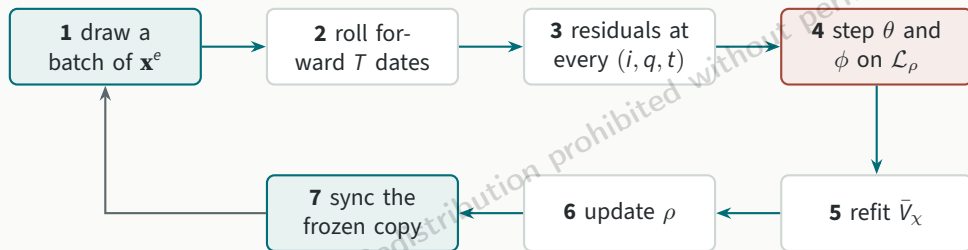


today's choices are trained;
tomorrow's rule, saving and
value are read from the copy

- ▶ The frozen copy supplies $\bar{\tau}^k(\mathbf{x}_+)$ and $\bar{g}$ inside ϑ_1 , and $\bar{V}(\mathbf{x}_T)$ at the end of the rollout.
- ▶ Inside one step these perceptions do not move: they are targets, not parameters.
- ▶ Copying the live networks into the frozen ones imposes consistency: perceived = actual.
- ▶ It is Session 3's target network, now used to reach a fixed point in rules.
- ▶ **At convergence the rule the government expects of its successors is the rule it plays.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Session 3, *Bridge (3): the Household's Actor-Critic*.

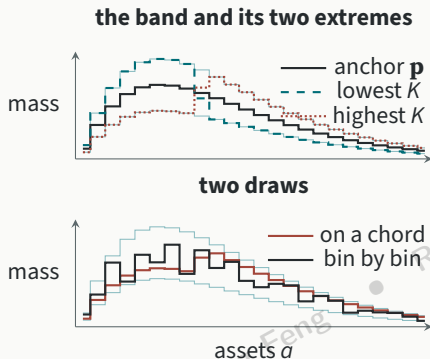
The Loop in One Picture



- Steps 1–4 run four times per iteration, each on a fresh batch; steps 5–7 once, or at longer intervals.
- Steps 2–4 are one forward and one backward pass through $T = 30$ matrices.
- **No outer loop: the fixed point in rules is reached by training and synchronising.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

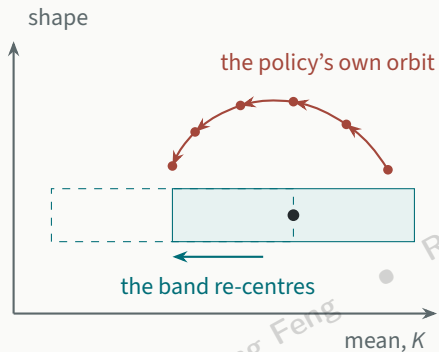
Step 1a: Sampling Distributions From a Band



- ▶ Around an anchor $\mathbf{p}$, each bin may move by a share ν , $p_i(1-\nu) \leq x_i \leq p_i(1+\nu)$ (thin lines); the total stays one.
- ▶ **Chords**: free mass to the richest, or poorest, bins gives the highest, or lowest, K . Draw K uniformly on the line from $\mathbf{p}$ to that extreme.
- ▶ **Bin by bin**: each x_i uniform in its interval, projected back to mass one: every shape, little range in K .
- ▶ **Chords give reach in K , bin-by-bin draws give shape: the rule learns only where it is sampled.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic: an illustrative anchor on 24 bins, $\nu = 0.35$.

Step 1b: The Band Follows, and the Policy's Own Orbit



- ▶ $\mathbf{x}^e, e = 1, \dots, E$: a batch of starting histograms, one rollout each. Half come from the band (Step 1a).
- ▶ Anchor $p_i \propto x_i^* e^{\kappa \bar{a}_i}$, moved every 10 iterations halfway to the classical K^* at current taxes: a taxing government moves both.
- ▶ Half from an **orbit**: every 20 iterations, $\mathbf{x} \leftarrow \mathbf{x} \hat{\Phi}(\mathbf{x}; \theta)$ for 60 steps, no gradient, plus the rule's own stationary $\mathbf{x}$.
- ▶ **The band covers off the path; the orbit is accurate on it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 2: Roll Forward T Dates

From one sampled $\mathbf{x}_0 = \mathbf{x}^e$, for $t = 0, \dots, T - 1$

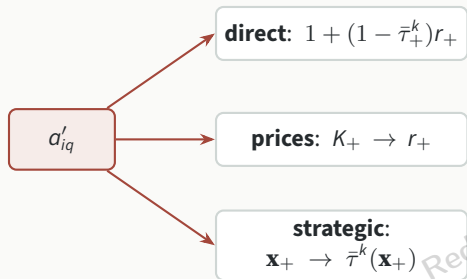
- 1 $K_t = \sum_i x_{t,i} \bar{a}_i$; taxes $(\tau_t^k, \tau_t^l) = \tau_\phi(\mathbf{x}_t)$; hours l_q , then L_t, r_t, w_t : no price loop
- 2 saving $a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$; consumption $c_{t,iq}$ from the budget; $y_{t,iq}$
- 3 revenue $G_t = \sum_{i,q} x_{t,i} p_q [\tau_t^k r_t \bar{a}_i + \tau_t^l w_t \varepsilon_q l_q]$
- 4 flow welfare $\bar{U}_t = \sum_{i,q} x_{t,i} p_q [u(y_{t,iq}) + \gamma_g \log(G_t + 10^{-3})]$
- 5 build $\hat{\Phi}_t$ from the $a'_{t,iq}$ by Deck 4.1's lottery; $\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}_t$, kept in the graph

$$W(\mathbf{x}^e; \theta, \phi) = \sum_{t=0}^{T-1} \beta^t \bar{U}_t + \beta^T \bar{V}(\mathbf{x}_T), \quad \bar{V} \text{ the frozen copy, a constant.}$$

- **Deck 4.1's rollout, with a tax at every date, the public good in welfare, and the matrix left in the graph.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $T = 30$; utility floored far below any value reached, to guard the logarithm and the power.

Step 3: The Constraint at Every (i, q, t) , and Its Three Channels



each moves ϑ_1 ; all three stay in the graph

- ▶ At each (t, i, q) and batch member: $\bar{E}_{t,iq} = \text{FB}(\vartheta_1, \vartheta_2)/u'(y_{t,iq})$.
- ▶ Inside ϑ_1 : $\mathbf{x}_+$ from the *live* saving rule; $\bar{\tau}$ and $\bar{g}$ from the frozen copy at $\mathbf{x}_+$.
- ▶ Deck 4.1 held a household's continuation fixed: it is its own marginal value (the envelope theorem).
- ▶ Here ϑ_1 is a *constraint*: no envelope theorem, and dropping a channel stalls the penalty.
- ▶ **The strategic channel, $\mathbf{x}_+ \rightarrow \bar{\tau}^k(\mathbf{x}_+)$, is the one no household sees.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 4: The Government's Loss

$$\mathcal{L}_\rho(\theta, \phi) = -W + \sum_{t,i,q} \beta^t x_{t,i} p_q \left[\mu_{t,iq} \bar{E}_{t,iq} + \frac{\rho}{2} \bar{E}_{t,iq}^2 \right] + B_{\text{feas}} + B_\sigma(\mathbf{x}^e) + \kappa_{\text{band}} B_{\text{band}}$$

- ▶ An augmented Lagrangian, multipliers μ . **As run, $\mu \equiv 0$: a pure quadratic penalty, weight ρ .**
- ▶ Weights $\beta^t x_{t,i} p_q$ price each residual by its mass; B_{feas} , same weights: $[(-c)_+ + (-y)_+] / 10^{-4}$.
- ▶ $B_{\text{band}} = \sum_t \beta^t \sum_i [(p_i(1 - \nu) - x_{t,i})_+ + (x_{t,i} - p_i(1 + \nu))_+]$, $\kappa_{\text{band}} = 20$: rollouts stay in the trained region.
- ▶ $B_\sigma = \sum_\tau [(0.02 - s_\tau)_+^2 + (s_\tau - 0.98)_+^2]$, $s_\tau = \tau / \tau_{\text{max}}$, at the sampled $\mathbf{x}^e$: zero inside, it keeps the sigmoid's slope alive, so the rule cannot freeze at its cap.
- ▶ No gradient through the weights, the $u'(y)$ dividing $\bar{E}$, or $\bar{V}(\mathbf{x}_T)$; one backward pass, one Adam step on θ and one on ϕ .
- ▶ **Every term but $-W$ is a solver device; the economics is $-W$ and $\bar{E} = 0$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Nocedal & Wright, *Numerical Optimization* (2006), ch. 17.

How a Network Responds to the Distribution

- ▶ The N bin masses enter the first layer like any other input. For the tax rule, two hidden ReLU layers:

$$\tau_{\phi}^k(\mathbf{x}) = \tau_{\max}^k \sigma(\ell_{\phi}(\mathbf{x})), \quad \ell_{\phi}(\mathbf{x}) = w_3^{\top} \text{ReLU}(W_2 \text{ReLU}(W_1 \mathbf{x}^{\top} + b_1) + b_2) + b_3.$$

- ▶ Its response to the mass in bin i , with D_1, D_2 the diagonal masks of active units:

$$\frac{\partial \tau_{\phi}^k}{\partial x_i} = \tau_{\max}^k \sigma'(\ell_{\phi}) [w_3^{\top} D_2 W_2 D_1 W_1]_i.$$

- ▶ Mass sums to one, so only moves count: δ from bin j to bin i changes τ^k by $\delta (\partial_i \tau^k - \partial_j \tau^k)$. A rise in K is one such direction; a mean-preserving spread is another.
- ▶ The saving rule $g_{\theta}(\varepsilon, \bar{a}_i; \mathbf{x})$ and the value $\bar{V}_{\chi}(\mathbf{x})$ read $\mathbf{x}$ the same way.
- ▶ $\partial \tau / \partial \mathbf{x}$ is learned, not imposed: it is how the government answers inequality, and it enters every saving update.

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Two hidden layers of 256 units; τ^l has its own output ℓ .

Updating the Tax Rule, One Date at a Time

$$\nabla_{\phi} \mathcal{L}_{\rho} = \sum_{t=0}^{T-1} \left[\frac{\partial \mathcal{L}_{\rho}}{\partial \tau_t^k} \nabla_{\phi} \tau_{\phi}^k(\mathbf{x}_t) + \frac{\partial \mathcal{L}_{\rho}}{\partial \tau_t^l} \nabla_{\phi} \tau_{\phi}^l(\mathbf{x}_t) \right] + \nabla_{\phi} B_{\sigma},$$

$$\frac{\partial \mathcal{L}_{\rho}}{\partial \tau_t^k} = -\beta^t r_t \left[\underbrace{\frac{\gamma_g K_t}{G_t + 10^{-3}}}_{\text{value of revenue}} - \underbrace{\sum_{i,q} x_{t,i} p_q u'(y_{t,iq}) \bar{a}_i}_{\text{cost to households}} \right] + \rho \sum_{i,q} \beta^t x_{t,i} p_q \bar{E}_{t,iq} \frac{\partial \bar{E}_{t,iq}}{\partial \tau_t^k}.$$

- g_{θ} does not read τ , so $\mathbf{x}_t$ does not depend on ϕ : each term sits at one date, at the $\mathbf{x}_t$ reached.
- The bracket is the flow trade-off, node by node; the penalty enters through today's $u'(y)$ in ϑ_1 .
- **Adam pushes τ_{ϕ} toward this condition at every $\mathbf{x}$ visited: the rule learns its response to the distribution only where the sampler and rollouts take it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. r net; $\partial \bar{E} / \partial \tau_t^k$ through $u'(y_{t,iq})$, the divisor held fixed. τ^l : the same form, plus terms through hours, L , r , w .

Updating the Saving Rule: Through the Law of Motion

$$\mathbf{x}_{t+1} = \mathbf{x}_t \hat{\Phi}_t, \quad \hat{\Phi}_t = \hat{\Phi}(a'_t), \quad a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t), \quad \frac{d\mathbf{x}_0}{d\theta} = 0.$$

$$\frac{d\mathbf{x}_{t+1}}{d\theta} = \underbrace{\frac{d\mathbf{x}_t}{d\theta} \hat{\Phi}_t}_{\text{displaced mass, carried on}} + \underbrace{\mathbf{x}_t \frac{\partial \hat{\Phi}_t}{\partial \theta}}_{\text{where today's mass lands}} + \underbrace{\mathbf{x}_t \frac{\partial \hat{\Phi}_t}{\partial \mathbf{x}_t} \frac{d\mathbf{x}_t}{d\theta}}_{\text{the saving rule reads } \mathbf{x}_t}$$

- Product rule on $\mathbf{x}_t \hat{\Phi}_t$, then chain rule: $\hat{\Phi}_t$ moves with θ directly and through $\mathbf{x}_t$.

Parameters	The gradient reaches them through
saving θ	consumption; the three terms, date after date; the taxes at each $\mathbf{x}_t$; the channels of ϑ_1
tax rule ϕ	the budget; G in welfare; $u'(y)$ in $\bar{E}$. Never $\mathbf{x}_+$: g_θ does not read τ
neither	$\bar{V}(\mathbf{x}_T)$; the frozen copy, a constant

- **Through θ , the gradient runs back along every $\hat{\Phi}_t$; through ϕ , only within the date.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\partial \hat{\Phi} / \partial a'$ is the lottery's slope, $\pm 1 / (\bar{a}_{j+1} - \bar{a}_j)$, zero where the clamp binds; the third term contracts over $\mathbf{x}_t$. Autograd runs the same chain in reverse.

Updating the Saving Rule: What the Distribution Sends Back

$$\lambda_t \equiv \frac{d\mathcal{L}_\rho}{d\mathbf{x}_t} = \frac{\partial \ell_t}{\partial \mathbf{x}_t} + \left(\lambda_{t+1} + \frac{\partial \ell_t}{\partial \mathbf{x}_{t+1}} \right) J_t, \quad J_t = \frac{\partial \mathbf{x}_{t+1}}{\partial \mathbf{x}_t}, \quad \lambda_T = 0,$$

$$\nabla_\theta \mathcal{L}_\rho = \sum_{t=0}^{T-1} \left[\frac{\partial \ell_t}{\partial \theta} + \left(\lambda_{t+1} + \frac{\partial \ell_t}{\partial \mathbf{x}_{t+1}} \right) \mathbf{x}_t \frac{\partial \hat{\Phi}_t}{\partial \theta} \right], \quad \ell_t : \text{date } t\text{'s terms of } \mathcal{L}_\rho.$$

In $\partial \ell_t / \partial \mathbf{x}_t$ who sits where, $\sum_q p_q U_{iq}$; prices via K_t ; **the tax rule's answer** $\partial \tau_\phi / \partial \mathbf{x}_t$; g_θ 's input
In $\partial \ell_t / \partial \mathbf{x}_{t+1}$ in $\bar{E}_t$: $\bar{\tau}^k(\mathbf{x}_{t+1})$, $r(\mathbf{x}_{t+1})$, $\bar{g}(\cdot; \mathbf{x}_{t+1})$: the strategic channel

- ▶ The tax rule's answer reaches θ twice: through $\tau_\phi(\mathbf{x}_t)$ along the path, and through $\bar{\tau}(\mathbf{x}_{t+1})$ in each Euler residual.
- ▶ **The saving rule is updated knowing how the tax will answer the distribution its own saving creates.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. The previous page run in reverse: one backward pass, $d\mathbf{x}_t/d\theta$ never formed. Penalty weights held fixed: "who sits where" enters through welfare only; $\bar{V}(\mathbf{x}_T)$ sends nothing back.

What Is New Here

	the state	who acts on $\mathbf{x}$	the gradient through $\mathbf{x}$'s motion
Krusell–Smith; Deck 4.2	mean, moments, histogram, shocks	nobody: households take it as given	one period, or a household panel
Markov taxes, one agent	K	a government	no distribution
HA taxes, commitment	a given $\mathbf{x}_0$	a government, once	a path of taxes, not a rule
this deck	the whole histogram	a government, every date	T dates, with $\partial\tau/\partial\mathbf{x}$ inside

- ▶ Without commitment the tax is a function $\tau(\mathbf{x})$; with commitment, a path chosen once.
- ▶ **To the authors' knowledge, the first time-consistent tax rule computed on the whole distribution as the state.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. One agent: Klein, Krusell & Ríos-Rull (2008). Commitment: Bhandari et al. (2021); Dyrda & Pedroni (2022).

Steps 5 and 7: Refit the Value Beyond the Horizon; Sync

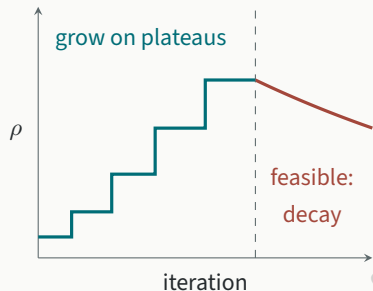
- **Targets**, every 20 iterations, at fresh band draws $\mathbf{x}$. As run, flow welfare held forever:

$$\hat{V}(\mathbf{x}) = \frac{\bar{U}(\mathbf{x})}{1 - \beta}, \quad \bar{U}(\mathbf{x}) = \sum_{i,q} x_i p_q U_{iq}.$$

- **Fit**, every iteration: 20 Adam steps on $\text{mean}[\bar{V}_\chi(\mathbf{x}) - \hat{V}(\mathbf{x})]^2$, batches of 256, after a long prefit.
- **Sync**, every 10 iterations: $g_\theta, \tau_\phi, \bar{V}_\chi$ copied into $\bar{g}, \bar{\tau}, \bar{V}$. A hard copy, not a moving average.
- Beyond T the economy is valued as if it stayed where the rollout ended; at $\beta^T \approx 0.2$ that term is not negligible.
- **$\bar{V}$ supplies a level at the horizon, never a gradient; the rollout supplies the gradient.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 6: The Penalty Weight Follows the Residual



- ▶ Each iteration, read $\bar{E}^{\text{test}}$: the mass-weighted mean of $|\bar{E}|$ on a fixed test batch.
- ▶ Grow, $\rho \leftarrow \min(1.015\rho, 10^4)$, if the loss stalls ($|\Delta\mathcal{L}/\mathcal{L}| \leq 1\%$) while $\bar{E}^{\text{test}} \geq \text{tol}$ and not improving.
- ▶ Decay, $\rho \leftarrow 0.99\rho$, once 5 of the last 10 reads are below tol.
- ▶ Learning rates scaled by $1/\sqrt{1 + 0.1\rho}$.
- ▶ **Measure tol at the solution: below what is reachable, ρ grows without end.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic curves: direction and shape only.

The Algorithm

Taxation without commitment, as a trained fixed point

- 1 warm-start g_θ from the no-tax solve; tax rule near zero; prefit $\bar{V}_\chi$; copy all to the frozen set
 - 2 four times: draw $\mathbf{x}^e$, half band, half orbit (step 1)
 - 3 roll forward T dates, $\hat{\Phi}_t$ in the graph; form W and every $\bar{E}_{t,iq}$ (steps 2–3)
 - 4 one backward pass of $\mathcal{L}_\rho$; one Adam step on θ , one on ϕ (step 4)
 - 5 refit $\bar{V}_\chi$; every 20 iterations new targets and orbit, every 10 a moved anchor (step 5)
 - 6 update ρ and the learning rates from the test-batch residual (step 6)
 - 7 every 10 iterations: frozen $\leftarrow$ live, “perceived = actual” (step 7)
 - 8 stop at the iteration budget, or early if the Euler error blows up; keep the best checkpoint
- **Each line is Deck 4.1’s machinery or Session 3’s target network. What is new is who owns the loss.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

What the Gradient Requires

Requirement	How it is met
a differentiable transition	lottery weights piecewise linear in a' (Deck 4.1); kinks of measure zero
no sampling noise	quadrature over ε , every asset node: a deterministic loss
a gradient in the tax rule	the barrier keeps the sigmoid off 0 and 1
a gradient at the limit	$\vartheta_2 = a'$ and the smoothed root keep constrained nodes live
memory for the graph	T dates of $N \times Q$ evaluations, $N \times N$ matrices, next-date copies in ϑ_1

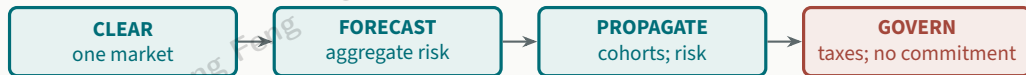
- ▶ T is a budget. Beyond it, $\bar{V}$ adds a level and no gradient.
- ▶ Lab block B: derivatives through up to twenty lottery steps match finite differences to 2×10^{-10} .
- ▶ **Every row is a place where a gradient can vanish without an error message.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the lab, block B.

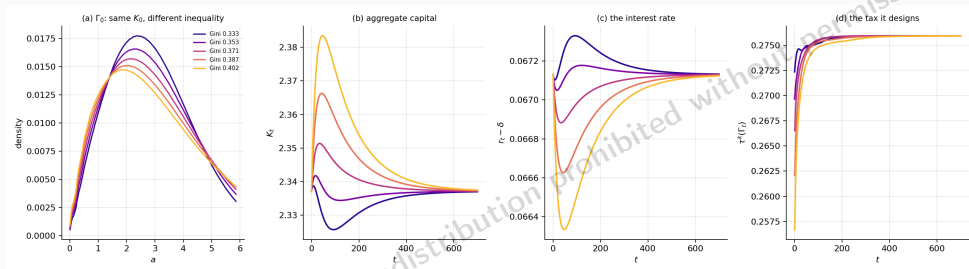
Session 4.4

What the Rule Does Along a Path

Start the trained economy from different distributions and let the tax rule, saving and prices run



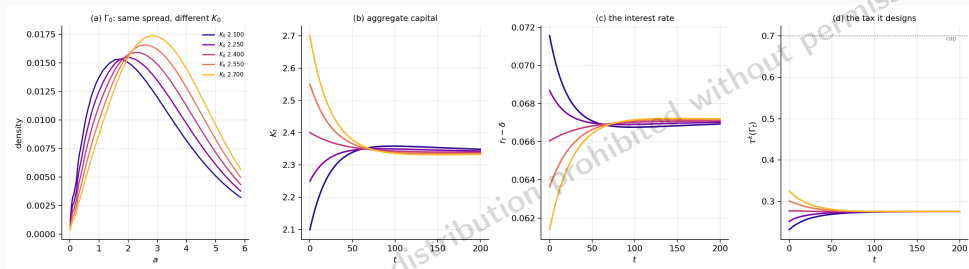
At the Same Capital, Inequality Alone Moves the Tax



- ▶ Five starts Γ_0 share one K_0 and differ only in inequality (a). At date 0, hours, prices and both tax bases are identical.
- ▶ The more unequal start is taxed less on impact (d); its capital overshoots further before returning (b, c).
- ▶ Every start converges to the same Γ^* and the same long-run tax.
- ▶ **A property of the trained rule: why it responds to shape is not yet settled.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Simulated with the trained Stage 2A networks.

At the Same Inequality, More Capital Means a Bigger Opening Levy



- ▶ The mirror experiment: the spread is held and K_0 moves (a). Now the tax base moves too, and the tax responds far more (d).
- ▶ The levy is transitional: capital and the interest rate return (b, c), and every start reaches the same long-run rate.
- ▶ A cap that never binds at the rest point can still censor the path: size it for the levy.
- ▶ **Without commitment: tax the capital inherited, then settle down.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Simulated with the trained Stage 2A networks.

Bridge: the Lab, and Session 5

Lab, Part I: the classical benchmarks of Decks 4.1–4.3.

Blocks D, E: Krusell–Smith by Deep Equilibrium Nets and in the sequence space.

Block F: the OLG economy, a market-clearing layer and a homotopy.

This deck: a government that differentiates through Γ .

- ▶ Cost: GPU-hours, not seconds. Stage 1B alone took about 2.9 GPU-hours, against 30 CPU-seconds for Stage 1A.
- ▶ It buys rules on the whole distribution, and the path from any start in the sampled region.
- ▶ Next in the research: aggregate shocks, $\hat{\Phi}$ conditioned on (z, z') ; then persistent ε .
- ▶ **The lab trains the Deck 4.2 and 4.3 networks in PyTorch, each graded by a gate fixed before the run.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the Session 4 lab.